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Chronic Pain

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SUMMARY PEARLS

- Chronic pain results in reduced physical functioning, decreased ability to make a living, and reduced societal contributions that can exacerbate stressors and affect mental health, resulting in a lower quality of life.
- The pain experience involves both physiologic and psychosocial responses, making treatment in a multidisciplinary pain center using the biopsychosocial approach optimal for assessment/treatment planning.
- Cognitive behavioral therapy for chronic pain is effective at restoring function, improving mood, and reducing pain interference among individuals with disabling chronic pain.
- Although opioids have been shown to be effective for pain control, long-term opioid therapy for chronic noncancer pain is generally discouraged.
- Buprenorphine may be helpful in the management of individuals with chronic pain, especially those with substance use disorder, utilizing cannabis/CBD/alcohol, or with concomitant benzodiazepine prescriptions.
- Long-term management should be individualized with combination treatment such as exercise/movement-based therapies, nonopioid pain medications (e.g., adjuvants), interventional modalities, and cognitive behavioral therapies for pain/cooccurring mental health diagnoses.

Introduction

Pain is a complex, dynamic, and subjective experience shaped by one's environment, learning, behaviors, and emotions. The International Association for the Study of Pain (IASP) defines pain as "an unpleasant sensory and emotional experience associated with, or resembling that associated with, actual or potential tissue damage."¹⁷² Acute pain serves an evolutionary protective purpose as it alerts to impending or actual harm and resolves with healing, whereas chronic pain is considered a disease state hallmarked by maladaptive behaviors, suffering, and disability, with often little correlation between underlying tissue injury and reported severity of pain. Pain is classified as chronic once it persists for 3 months or longer, exceeding the time of normal tissue healing.²⁴⁷

Impacts of Chronic Pain

The effects of chronic pain on an individual and society are extensive. In 2011, the economic impact of pain in the United States was estimated to be between \$560 and \$635 billion annually, encompassing medical costs, lost productivity, and disability. The economic cost of pain exceeds that of the nation's priority health conditions, such as cancer and heart disease.⁹⁵ The devastating impact of the opioid epidemic highlights the need for effective treatments for chronic pain. Between 2000 and 2014, the total number of deaths linked to the use of prescribed and illicit opioids surged by 400%.²¹⁶ In 2019, almost 50,000 individuals in the United States lost their lives to opioid-related overdoses.¹⁸⁵

Disability

Chronic pain can impact nearly every facet of a person's life, resulting in a lower quality of life, reduced physical functioning, and poor mental health. In a 2019 Global Burden of Disease Study, low back pain ranked fourth in the top 10 causes of disability-adjusted life-years, which represents the loss of equivalent years of full health.⁹⁸ Approximately half of those with chronic pain report work limitations, compared to just 15% of those without chronic pain.²⁶⁸

Prevalence

A 2021 National Health Interview Survey found the prevalence of chronic pain among US adults to be 20.9% (51.6 million people), with the most common pain locations involving the hip, knee, foot, and back.^{209,268} Prevalence rates in other industrialized countries are reported to range from 13% to 50%.^{114,176} Roughly 6.9% of US adults (17.1 million people) experience "high-impact" chronic pain, which is defined as chronic pain that significantly impairs daily activities.²⁰⁹ Chronic pain does not affect everyone equally. Higher rates are found in females, military veterans, non-Hispanic Indian or Alaska Native adults, adults identifying as bisexual, those who are divorced or separated, those from lower socioeconomic backgrounds, and residents of rural areas.^{64,209}

Pain Terminology

The IASP Task Force on Taxonomy defines neuropathic pain as "pain caused by a lesion or disease of the somatosensory nervous system" and nociceptive pain as "pain that arises from actual or

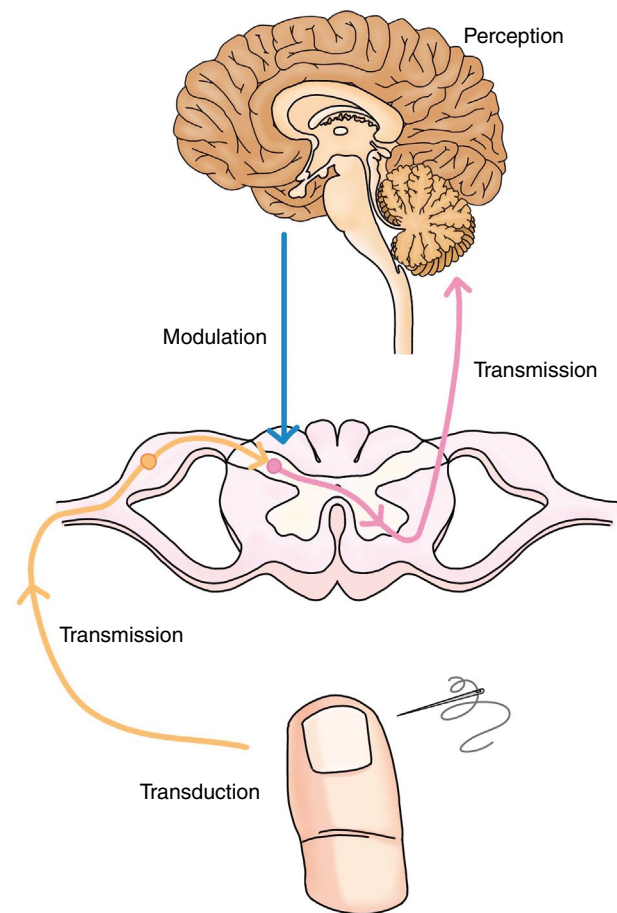
threatened damage to non-neural tissue and is due to the activation of nociceptors.”¹²⁶ Nociceptive pain occurs in the context of a normally functioning somatosensory nervous system, whereas abnormal function is seen in neuropathic pain. Neuropathic and nociceptive pain were the only two classes of pain until 2017, when the IASP adopted nociplastic pain, which is defined as “pain that arises from altered nociception despite no clear evidence of actual or threatened tissue damage causing the activation of peripheral nociceptors or evidence for disease or lesion of the somatosensory system causing pain.”¹²⁶ There are instances where an overlap of these pain types may be at play simultaneously.⁹⁴ See Table 38.1 for additional pain terminology.

Pain Physiology

The pain experience consists of signal transduction peripherally, transmission to the central nervous system, perception in the brain, and modulation at the spinal cord (Fig. 38.1).

Transduction

Transduction refers to the process by which a chemical, mechanical, or thermal stimulus is transformed into an electrical signal in primary afferent nerves. In regard to pain, this occurs when a stimulus capable of causing tissue damage activates nociceptors on free nerve endings of C and A-delta fibers. C fibers are polymodal, meaning they respond to a variety of stimuli (mechanical, chemical, and/or temperature), while most A-delta fibers respond to thermal and mechanical stimuli. In response to tissue injury,



• **Fig. 38.1** Pain physiology. The pain experience consists of signal transduction peripherally, transmission to the central nervous system, modulation at the spinal cord, and perception in the brain. (Illustration courtesy Sophia Duong.)

TABLE 38.1 International Association for the Study of Pain Terminology

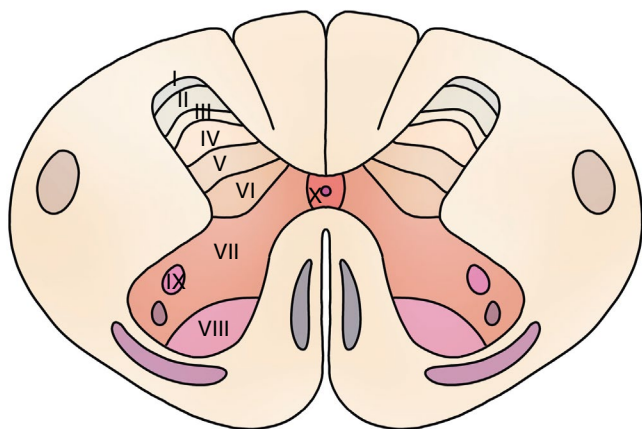
Terminology	Definition
Allodynia	Pain due to a stimulus that does not normally provoke pain
Dysesthesia	An unpleasant abnormal sensation, whether spontaneous or provoked
Paresthesia	An abnormal sensation, whether spontaneous or provoked
Hyperalgesia	Increased pain from a stimulus that normally provokes pain
Hyperesthesia	Increased sensitivity to stimulation
Hypoalgesia	Decreased pain in response to a normally painful stimulus
Hypoesthesia	Decreased sensitivity to stimulation
Analgesia	Absence of pain in response to stimulation that would normally be painful
Neuralgia	Pain in the distribution of the nerve(s)
Neuritis	Inflammation of a nerve or nerves
Nociception	The neural process of encoding noxious stimuli
Sensitization	Increased responsiveness of nociceptive neurons to their normal input, and/or recruitment of a response to normally subthreshold inputs

From Merskey H, Bogduk N. *Classification of Chronic Pain*. 2nd ed. International Association for the Study of Pain Task Force on Taxonomy. IASP Press; 1994.

byproducts from cellular damage (potassium ions, histamine, and prostaglandins) and platelet activation (serotonin) bind to and activate specific channels, thereby increasing membrane permeability and contributing to depolarization of the nerve terminal. Mechanical stimuli generate an action potential by augmenting receptor ion permeability through deformation of the receptor. A stimulus of sufficient magnitude is required to reach the depolarization threshold in the primary afferent nerve terminal to generate an action potential. The action potential frequency is directly proportional to the intensity of the stimulus as higher receptor stimulation correlates to increased channel opening.

Transmission

Transmission refers to the process by which a nociceptive message is transmitted from the periphery to the central nervous system. The energy created from a noxious stimulus is converted to an action potential, which is then propagated by voltage-gated sodium channels (NaV 1.7, NaV 1.8) along the primary afferent nerve. These afferent nerves synapse on second-order neurons in the dorsal horn of the spinal cord, which is divided into 10 different laminae (Fig. 38.2). The C fibers synapse on second-order neurons in lamina I (marginal zone), lamina II (substantia gelatinosa), and lamina X (central canal). The A-delta fibers also synapse in lamina



• **Fig. 38.2** Laminae of the spinal cord. (Illustration courtesy Sophia Duong.)

I (marginal zone) and lamina II (substantia gelatinosa), and additionally they synapse in lamina V in the nucleus proprius. Innocuous sensory information (touch, proprioception) is carried to deeper zones of the dorsal horn (nucleus proprius or laminae III–VI) via low threshold, large, myelinated afferents (A-beta fibers) and continue on through the dorsal column and medial lemniscus tracts to the somatosensory cortex. Painful and non-painful stimuli converge in the nucleus proprius and synapse on wide dynamic range (WDR) neurons in lamina V.

The majority of pain signals arise from laminae I, II, and V and travel to the brain via second-order neurons in the neospinothalamic tract, which is responsible for sensory-discriminative aspects of pain (location and intensity of stimulus).²³ The neospinothalamic tract (phylogenetically newer) decussates in the anterior white commissure of the spinal cord before ascending to the thalamus on the opposite side from entry and then to the somatosensory cortices. Some nociceptive signals decussate and ascend via the paleospinothalamic (phylogenetically older), spinoreticular, and spinomesencephalic tracts. These tracts are involved in the motivational-affective aspects of pain, poorly localized pains, and descending modulation through connections to the anterior cingulate cortex, prefrontal cortex, and insula.

Myelinated A-delta fibers are involved in the first or quickest phase of pain perception because of their faster conduction velocity; C fibers are responsible for slower onset, poorly localized pain. Take, for example, when a toe collides with a bed post. There is an immediate sharp pain felt precisely in the injured toe (A-delta fibers) followed by a wave of diffuse pain that builds over the course of several seconds (C fibers). The fast, sharp pain mediated through A-delta fibers serves a protective role that allows one to quickly localize and respond to a noxious stimulus, and the slower, more diffuse, persisting pain serves as a reminder to be cautious while the injury heals. There are mutations that alter these evolutionary protective mechanisms. Loss-of-function mutations in the genes encoding for NaV 1.7 channel result in congenital insensitivity to pain,⁶⁰ while gain-of-function mutations cause conditions such as small-fiber neuropathies and inherited erythromelalgia (Table 38.2).²⁶

Perception

Nociception describes the unconscious processing of noxious stimuli in the central and peripheral nervous systems, whereas perception of pain is the conscious awareness and subjective experience of noxious sensations involving attention, interpretation, and expectation. The

TABLE 38.2 Types of Afferent Fibers

Nerve Fiber Group	Conduction Velocity	Axon Diameter	Stimuli/Effect
A-beta	30–90 m/sec	5–15 mm	Tactile sensation, proprioception
A-delta	6–30 m/sec	1–5 mm	Thermal, mechanical stimuli Fast, highly localized pain
C	0.5–2 m/sec	0.5–2 mm	Thermal, mechanical, and chemical stimuli Slow, poorly localized pain

neural circuitry responsible for pain perception is incompletely understood, but it is believed to involve thalamocortical and corticoclimbic structures.¹⁶⁸ The relationship between perception of pain and injury is not always linear. Pain perception may exist in the absence of objective physical injury (i.e., functional pain disorders), and nociception can occur without the awareness of pain (i.e., a person in shock who appears to be seemingly oblivious to significant injuries just sustained in a traumatic accident).

There are many locations within the brain that are involved in pain processing, collectively termed the neuromatrix. The primary and secondary somatosensory cortices are responsible for sensory-discriminative aspects of pain, whereas the anterior cingulate cortex, prefrontal cortex, and insula are involved in the emotional, motivational, and subjective aspects of pain.¹⁴

Modulation

Descending modulation involves several cortical and subcortical regions that exert a facilitatory or inhibitory effect on pain at the level of the dorsal horn through the release of monoamines, and this can impact the pain experience. The periaqueductal gray matter (PAG) is a key structure involved in descending modulation as it receives input from several supraspinal structures, such as the cortex, amygdala, and hypothalamus, and has bidirectional communication with the rostral ventromedial medulla (RVM), which then projects onto the dorsal horn.^{25,49,122} There are spinal receptor subtypes in the dorsal horn that dictate the pronociceptive or antinociceptive response to norepinephrine, dopamine, and serotonin. Analgesic effects are mediated by inhibitory serotonergic, norepinephrine, mu-opioid, enkephalin, and gamma-aminobutyric acid (GABA) pathways.

The endogenous opioid system exerts an antinociceptive effect through the activation of descending norepinephrine pathways. Opioid receptors are found throughout the descending pain modulatory pathway. Supraspinal mechanisms involve the PAG, which activates descending pathways and affects the motivational-affective centers through connections to the forebrain via projections from the dorsal raphe nucleus (serotonin) and the locus coeruleus (norepinephrine). In the spinal cord, opioids block the presynaptic release of transmitters from primary afferents and hyperpolarize second-order neurons. Damage to the RVM or amygdala has been found to interrupt analgesic effects of morphine.^{90,163}

There are additional spinal mechanisms that modulate pain. The substantia gelatinosa (lamina II) contains numerous local interneurons that connect more superficial dorsal horn layers to deeper layers (laminae III–VI) and regulate local excitability. WDR neurons, located in lamina II and more so in lamina V,

receive input from a range of stimuli, including high-threshold A-delta and C fibers (noxious), low-threshold A-beta fibers (touch, proprioception), and visceral afferents. Input from A-delta and C fibers inhibit inhibitory WDRs and “open the gate” for pain transmission to continue to the brain. Concurrent input from touch/pressure via A-beta fibers activates inhibitory WDRs thereby “closing the gate” and reducing activation of the ascending secondary neuron (Fig. 38.3). This gating mechanism of WDR is responsible for the analgesic effects of rubbing a painful area, cryo- and thermotherapy, transcutaneous electrical stimulation, and spinal cord stimulation. Descending modulation from cortical areas can also close the gate via release of endorphins, which is thought to be the mechanism behind why a positive mindset, meditation, and exercise can improve pain. Conversely, negative emotions can open the gate and worsen the pain via descending modulation.

Central Sensitization

Wind-up phenomenon occurs in the dorsal horn when there is repetitive stimulation of C fibers from prolonged pain states. This results in heightened pain intensity, increased receptive field of pain, and persistent pain even after the stimulus is removed. Wind-up is mediated through the activation of N-methyl-D-aspartate (NMDA) receptors, protein kinases, descending facilitatory serotonergic pathways, astrocytes, microglia and release of prostaglandins, and nitric oxide.

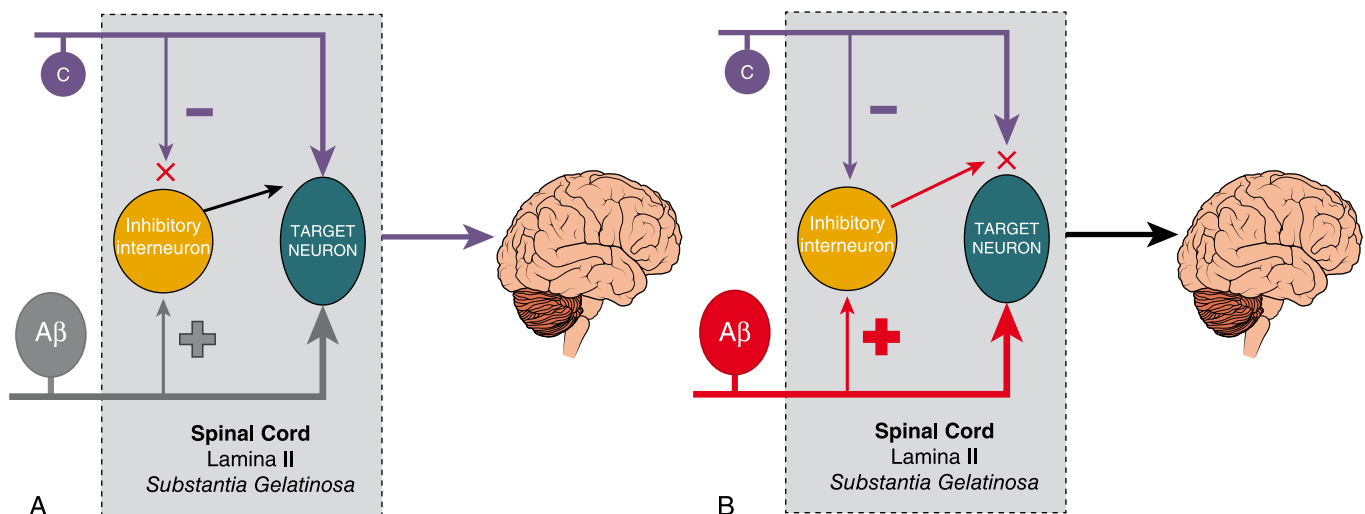
Peripheral Sensitization

In the periphery, pain sensitization occurs when the primary afferent nociceptive fibers are provoked with a much lower stimulus or become spontaneously active leading to hyperalgesia and allodynia. Changes in afferent fiber receptive fields can also be seen.^{111,244} This is mediated through release of various substances in response to tissue trauma, inflammation or infection such as amines, bradykinin, prostaglandins, cytokines, substance P, calcitonin gene-related peptide (CGRP), proteinases, potassium ions, and protons.^{40,54,85,130,155,174} These inflammatory substances

trigger further activation of primary afferent fibers in a dose-dependent fashion.³⁰ Additionally, depolarization of nociceptive afferent terminals results in activation of protein kinases, which alters and increases the number of ion channels (voltage-gated sodium channels and transient receptor potential vanilloid-1 [TRPV-1] channels) at the nerve terminals contributing to the pathologic hyperexcitability of the nociceptors.⁸⁶ The release of neuropeptides, CGRP, and substance P peripherally cause vasodilation and extravasation of plasma proteins (neurogenic inflammation), which acts as a feed-forward loop of sensitization.¹⁵⁸

Pain Theory

There have been numerous pain theories and theoretic frameworks presented to encapsulate the pain experience. Several will be highlighted here. The specificity theory of pain refers to the idea that various sensations (pain, touch, temperature) are transmitted along distinct pathways. Rene Descartes (1596–1650) was one of the earliest Western philosophers to propose this theory, who also made the distinction between the transduction of pain signals in the periphery and pain perception in the brain.⁷³ Schiff, in 1858, confirmed this in animal models with experimental lesions of the spinal cord.²²¹ The concept that each sensory modality has specific receptors was founded by 19th-century contributions of von Frey, Blix, and Goldscheider.^{35,93,104} A revolutionary concept, known as the gate-control theory, was proposed in 1965 by Melzack and Wall, which posits that nonpainful stimuli can modulate and reduce the perception of pain via a gating mechanism in the dorsal horn of the spinal cord.¹⁷⁰ This was the first theory to introduce the concept that pain experience is not simply the result of a linear process that begins with the stimulation of pain pathways in the peripheral nervous system and ends with the experience of pain in the central nervous system. Rather, neural impulses that potentially signal pain from the peripheral nervous system are subject to a number of modulations in the spinal cord by a gatelike mechanism in the dorsal horn before the experience of pain is transmitted to the central nervous system. It also proposes that the gate mechanism is modulated by



• **Fig. 38.3** Gate-control theory. (A) Pain signals (via C and A-delta fibers) are transmitted to the brain unchecked after turning off the inhibitory interneuron and synapsing with the secondary target neurons in lamina II of the spinal cord. (B) When A-beta (Aβ) fibers are activated, they turn on the inhibitory interneurons to block (or attenuate) signals from C and A-delta fibers to the secondary target neurons, thereby closing the gate of pain transmission to the brain.

emotions, cognitive state, and past experiences. Although this theory is based on physiology, it explains both sensory and psychological aspects of pain perception.¹⁷⁰

Multidimensionality of the Pain Experience

The present-day understanding of pain involves many dimensions and behaviors that may ease, magnify, or prolong the pain experience. Melzack and Casey described how the sensory-discriminative (location, quality, duration, and intensity), affective-motivational (unpleasantness and associated sympathetic response), and cognitive-evaluative (context, cultural values, and cognitive status) dimensions of pain interact with each other.¹⁶⁹ It is apparent that pain, especially chronic pain, can lead to feelings of anxiety, depression, and exasperation; however, evidence supports that negative mood and emotion can cause, perpetuate, and exacerbate pain.²⁵⁹ For example, interpersonal rejection has been found to downregulate pain attenuating pathways and increase activity in neural pain circuits.¹⁵⁰ The insula and PAG play vital roles in the affective experience of pain. Lesions of the insula have been found to remove the experience of unpleasantness associated with pain perception (pain asymbolia),²²² and noninvasive neuromodulation of the insula produces dissociable effects in test subjects.²⁴² Learned defensive or aversive pain behaviors and associated sympathetic input involve the PAG, which exerts inhibitory or facilitatory influences on the dorsal horn of the spinal cord.²⁵ Examples of cognitive modulation of the pain experience include hypnosis, allowing a person to be less bothered by a painful stimulus of unchanged intensity,^{146,261} placebo and nocebo response,²⁵⁵ and cognitive behavioral therapy (CBT).^{218,219}

Cerebral Plasticity: Transition From Acute to Chronic Pain/Pain Chronification

The exact impetus of what causes pain to persist beyond the natural course of healing is unknown, but several physiologic differences are found in those experiencing chronic pain. These include pain-induced altered brain connectivity, cortical reorganization, maladaptive neuroplasticity, and gray and white matter loss. Those in chronic pain show less activation of normal pain processing areas and increased prefrontal cortex activity on imaging, which is thought to correlate to more emotional and cognitive influences and suggests the presence of dysfunctional endogenous pain regulatory systems.^{14,20,28,131} Studies have found altered default mode networks,²¹ resting state attentional networks,⁵⁰ and thalamocortical connectivity⁵¹ in patients with chronic pain. There is evidence of primary somatosensory cortex reorganization in conditions such as phantom limb pain, complex regional pain syndrome, and carpal tunnel syndrome, all of which show reversal with successful treatment.^{84,161,243} Additionally, studies have found gray and white matter changes in pain processing brain regions of those with chronic pain conditions.^{99,177,224} Gray matter loss was shown to be reversible after surgery.^{107,212}

Psychological Aspects of Pain

The fields of psychiatry and psychosomatic medicine progressed significantly in the 20th century. Sigmund Freud (1856–1939) emphasized the potential link between physical and psychological factors in numerous medical conditions. Later, disenchantment with Freud's psychoanalytic principles led to the development of the field of psychosomatic medicine and the subsequent rapid development

of the fields of health psychology and behavioral medicine in the 1970s.⁹⁷ Physicians such as George Engel (1913–1999) challenged the biomedical model of disease, noting insufficiencies because it failed to include the social, psychological, and behavioral dimensions of illness. His classic 1959 article, “Psychogenic Pain and the Pain-Prone Patient,” discussed various contextual meanings of persistent pain and the importance of an individual's interpretation of pain. Sternbach, in 1966, argued that physiologic and affective perceptions of pain should be understood as learned responses under the control of environmental forces; he addressed psychophysiologic pain syndromes, including “stress-induced pain disorders.”²³⁹ Wilbert Fordyce^{92a} later proposed an operant conditioning model of chronic pain based on an ends approach of identifying and treating various pain behaviors. More recently, higher cognitive functioning in pain states (such as memory and emotive components) was embraced in the cognitive behavioral approach, led by health psychologists such as Dennis Turk and Frances Keefe, emphasizing the role of attributions, efficacy, personal control, and problem solving. Thoughts and beliefs can influence, and be influenced by, emotional and physiologic responses.²⁵⁰ This has contributed to the evolution of a more clinically pragmatic school of pain assessment and treatment: the biopsychosocial model that incorporates the physical, cognitive, affective, and behavioral components related to the ongoing pain experience. Biologic factors can initiate a physical disturbance, but psychosocial factors often influence pain perception, pain behavior, and the ongoing pain experience.²⁴⁹

Biopsychosocial Formulation

In the biopsychosocial formulations of patients, it is important to consider biologic, psychological, and social factors. Biologic factors include genetics, neurochemistry, physical health, or effects of medication. Psychological factors may encompass thoughts, perceptions, behaviors, and emotions, including irritability and anger. Social factors may include solicitous dynamics, punishing dynamics, and supportive distractors such as ignoring pain behaviors; these can play a role in patients' engagement in activity with pain present and their willingness to have pain without trying to avoid or control it.¹⁶⁴ Additionally, cognitive, learning, and affective factors may lead to more or less pain; therefore intercepting in various points of patterns encompassing the above will allow for patients to change how they think and consequently feel. This can lead to improved function and quality of life. Somatic symptoms of depression include change in appetite/weight, loss of energy, and sleep disturbance. Anxiety related to chronic pain is an important factor involved in maladaptive responses, behavioral interference, and affective distress. These may contribute to avoidance activities that serve to promote ongoing pain, physical deconditioning, and social isolation. Such symptoms may be cooccurring in those with chronic pain and correlate with inactivity, experience of greater pain level and consequent disability, reduced cognitive functioning, and decreased ability/willingness to engage in treatment or appreciate potential benefit.

Consideration for Cooccurring Mental Health Diagnoses

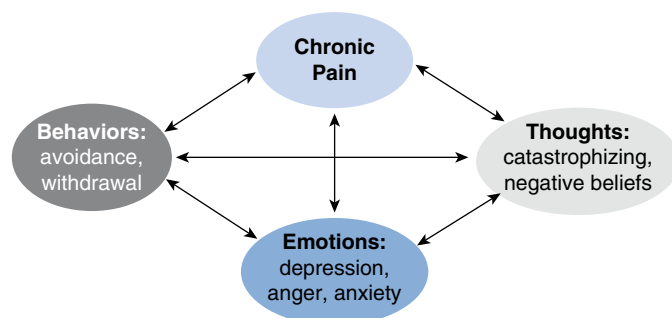
The importance of assessment/treatment of cooccurring medical and mental health conditions is significant when employing a biopsychosocial model. The fifth edition of the *Diagnostic and Statistical Manual of Mental Disorders* (DSM-5) made noteworthy

changes to previous taxonomy, and this updated terminology reflects a greater understanding of pain. In DSM-5, diagnostic criteria were modified; this included replacing *somatoform disorders* with *somatic symptom and related disorders* and eliminating more ubiquitous terms such as *chronic pain syndrome* and *psychogenic pain*. The new disorder somatic symptom disorder (SSD) and related disorders may be common to many patients presenting with chronic symptoms (persistent for at least 6 months) and is characterized by “somatic symptoms that are either very distressing or result in significant disruption of functioning, as well as excessive and disproportionate thoughts, feelings and behaviors regarding those symptoms.”¹¹ Such terminology is also reflected in the IASP’s updated definition of pain because it is vital to assess the patient’s pain/perception even in the absence of organic cause. A diagnosis of SSD with or without additional psychiatric diagnoses (i.e., depressive disorders, anxiety disorders) may be commonly seen in the psychiatric chronic pain encounter and, when recognized, may serve as an additional target for behavioral health interventions. Therapy, including CBT, may be helpful in such cases.

Cognitive Behavioral Therapy for Chronic Pain

Cognitive behavioral therapy for chronic pain (CBT-CP) is effective at improving function, mood, and pain interference among individuals with disabling chronic pain.¹⁰³ CBT-CP is focused on helping people with pain adopt strategies to improve their pain experience. Some of the strongest evidence for CBT-CP implementation has come from the Department of Veterans Affairs. CBT-CP has been shown to be an effective, safe treatment for veterans with chronic pain. This model can increase the availability of training in standardized, pain-focused, evidence-based, behavioral interventions (Fig. 38.4). Broad dissemination of such training would improve patient access to effective, nonpharmacologic treatment options in both the public and private sectors.¹⁸⁰

CBT-CP involves a structured approach that focuses on the relationships among cognitions (or thoughts), emotions (or feelings), and behaviors. Treatments based on cognitive behavioral theory have been successfully applied to the management of chronic pain, either delivered alone or as a component of an integrated, multimodal, and interdisciplinary pain management program. Sessions of CBT-CP include increased physical activity, time-based pacing, relaxation techniques, pleasant activity scheduling, sleep education, and cognitive coping techniques. Several will be reviewed in this chapter.



• **Fig. 38.4** The cognitive behavioral theory for chronic pain model, showing the interconnections between chronic pain, thoughts, emotions, and behaviors. (From Murphy JL, McKellar JD, Raffa SD, et al. *Cognitive Behavioral Therapy for Chronic Pain Among Veterans: Therapist Manual*. U.S. Department of Veterans Affairs; 2014.)

Time-based pacing is described as spending just enough time on an activity to get the most out of it without pushing so far that the individual ends up in a lot more pain, requiring time to recuperate, etc. This removes punishment from the activity and enables the individual to complete more in the long run. CBT-CP also integrates a walking program, in which walking is logged weekly. This exposure to physical activity is behaviorally activating. These strategies allow for accomplishing goals that reinforce being active. Several relaxation techniques reviewed during CBT-CP are deep breathing, progressive muscle relaxation, and guided imagery. With practice, elements of these modalities may be integrated into parts of daily life for improved stress/pain management.

Addressing various maladaptive beliefs, perceptions, and thoughts are fundamental within CBT. For example, fear-avoidance beliefs are significantly associated with the experience of pain, especially when the pain becomes chronic in nature. The anticipated threat of intense pain may commonly result in the constant vigilance and monitoring of pain sensations, which may lead to low-intensity sensations of pain becoming unbearable for the individual. Just the anticipation of increased pain or reinjury can further stimulate avoidance behaviors. A vicious cycle may develop in which fears of increased pain or reinjury contribute to the avoidance of many activities, leading to further inactivity and, ultimately, to greater disability.⁹⁶ Pain catastrophizing, a cognitive response style that views the experience of pain as uncontrollable, permanent, and destructive, has been identified as an important maladaptive cognition that contributes to difficulties with the management of chronic pain. Consequently, pain catastrophizing is commonly targeted in CBT for chronic pain.⁷⁷ Table 38.3 outlines some common unhelpful thoughts along with how to reframe these with patients.

Acceptance and commitment therapy (ACT) stemming from traditional CBT is an empirically based, third-wave CBT that uses acceptance, mindfulness, commitment, and behavior change strategies to increase psychological flexibility.^{63a} The ultimate goal in ACT for chronic pain is to clarify patients’ values so that they may improve their quality of life. Of note, there is a newer psychological therapy, called emotional awareness and expression therapy (EAET), showing promise for chronic pain. EAET is unique in that it targets problems, life stressors, or traumas that many individuals with chronic pain encounter, as centralized pain conditions appear to be strongly influenced by the functions of the central nervous system, including thoughts, feelings, and relationships. Research shows that stress throughout people’s lives, and how they handle the emotions from that stress, can trigger, worsen, or maintain their pain.^{6a} Most psychological treatments for pain conditions, including traditional CBT and ACT, do not focus on helping people resolve these earlier life stressors, but this is the focus of EAET. This form of therapy is rather new, and so only a handful of clinical trials have been published testing it. Research suggests that EAET can be quite powerful and lead to a significant reduction in pain.^{267a}

Sleep

The relationship between disturbed sleep and chronic pain conditions is well documented for both adults and adolescents.^{178,194} Patients with chronic pain may display frequent sleep fragmentation, longer sleep latency, and a decrease in overall quality of sleep. Sleep fragmentation is characterized by repetitive short interruptions in sleep and is a recognized factor in the cause of excessive daytime sleepiness. This inability to maintain sleep can be the most important factor in the treatment of disturbed sleep in individuals with chronic pain. The strength of this relationship between disturbed sleep, mood, and

TABLE 38.3 Common Pain Thoughts

Type of Thought	Common Pain Thoughts	More Helpful Thoughts
Catastrophizing	"This pain will never go away."	"I have had good days before, and I can have them again."
All-or-nothing thinking	"I can't do anything because of my pain."	"I can still do some things, even if I need to take it slow."
Self-blame	"I'm so weak for letting this pain get to me."	"Managing chronic pain is challenging, and I'm doing the best I can."
Hopelessness	"I can't enjoy my retirement because of pain."	"I can find new ways to enjoy retirement, even with this pain."
Helplessness	"There's nothing I can do to feel better."	"I can try different strategies to manage my pain."
Rumination	"I am suffering from pain every second."	"It's tough, but I can find ways to cope and move forward despite the pain."
Overwhelming thoughts	"I can't handle this anymore."	"I have handled difficult things in the past, and I can get through this."
Comparisons	"Everyone else is fine; why am I suffering?"	"Everyone has challenges, and I can work through mine."
"Should" statements	"I should be able to handle this better."	"It's okay to seek help and take it one step at a time."
"Should" statements	"I should not feel this pain."	"It's normal to feel pain sometimes, and I can focus on managing it."
Disqualifying the positive	"It doesn't matter that I felt okay yesterday, today is terrible."	"Good days show that I can have relief, and I will work towards more of them."
Jumping to conclusions	"This treatment won't work, so why bother?"	"It's worth trying different treatments to see what helps manage my pain."
Emotional reasoning	"I can't do anything about this pain, so I must be useless."	"Feeling this way doesn't mean it's true; I can still find ways to be productive and fulfilled."

chronic pain is significant. It is important to treat underlying sleep disorders, such as obstructive sleep apnea. Additionally, CBT for insomnia is a first-line treatment of chronic insomnia that may play a significant role in interdisciplinary team management.

Pain Assessment

In addition to comprehensive medical history, psychosocial history, and a physical exam, pain assessment is multifaceted and should include specific characteristics, such as location, intensity, quality, among others.

Pain Location

The location of pain helps narrow the diagnosis. While location of pain often points to the area of injury in musculoskeletal conditions, this may not apply in neuropathic or visceral pain. Pain from a nerve injury, such as radiculopathy, will present distally in the area of nerve innervation. Likewise, visceral pain arising from the heart or pancreas will refer to other areas of the body (arm and midback, respectively). The pain drawing, which consists of front and back drawings of the body, is a useful tool for patients to indicate the area of pain by shading in the location where they experience pain. Additionally, utilizing symbols drawn on the body can be used to convey other aspects of pain (i.e., xx-burning, 00 pins and needles, // stabbing, == numbness). Several pain instruments, such as the McGill Pain Questionnaire and the Brief Pain Inventory, incorporate the pain drawing in the assessment section.

Pain Intensity

Multiple approaches are available to rate pain intensity. These include the verbal rating scale, the visual analog scale, and the verbal categorization of pain (mild, moderate, severe). The numeric

rating scale is one of the most popular scales due to its ease of use. It asks patients to either express or write down their pain intensity using a numeric rating from 0 (no pain) to 10 (most intense pain imaginable). In pediatric patients or those with verbal challenges, the Wong-Baker faces scale can be used, where patients point to a picture of a facial expression that best reflects their pain.⁹¹

Pain Characteristics/Quality

The type of pain experienced by the patient can give clues to the diagnosis. Although there is significant overlap in the presentation of pain from different etiologies, musculoskeletal pain is often described as an aching, crampy, sharp pain often worse with movement or activities. Pain of a musculoskeletal nature is often worse when the structure is stressed by stretching, mechanical compression, poor position, overuse, among others. Neuropathic pain is often described as a burning, shooting, tingling sensation that is worse with positions/movements that cause further nerve injury or with provocative testing such as the Tinel test for nerve compression and straight-leg raise test for lumbosacral radiculopathy. Visceral pain can be vague, dull, and poorly localized, leading to pain in other locations (referred pain) and can be exacerbated when the affected organ is stressed.

Temporal Pattern

The onset of pain can be difficult to assess in chronic pain states, especially when the pain is not associated with an injury or event. Assessment of duration of pain is important for the classification of chronic pain (>3 months).¹²⁶ History of the temporal pattern and variability of the pain (intermittent, constant, etc.) with position (sitting, lying, standing) or activity (walking, driving, eating, etc.) is important in determining exacerbating factors. This may help with establishing the diagnosis as well as generating a management plan.

Quantitative Sensory Testing

Quantitative sensory testing (QST) involves assessing patients' perceptual responses to quantifiable sensory stimuli, which include thermal (heat, cold), mechanical (tactile, pressure, vibration), electrical, ischemic, and chemical modalities. By selectively engaging different nerve endings and pathways, QST helps evaluate somatosensory function and pain processing.¹⁸ Dynamic QST approaches also provide insights into pain inhibition and facilitation.⁶¹ Clinically, QST is most commonly used for characterizing neuropathic pain.⁹¹

Skin Biopsies

Epidermal nerve fiber density (ENFD), commonly measured via punch biopsies, is used to diagnose small-fiber peripheral neuropathy. Although recent evidence suggests that ENFD may not be directly related to neuropathic pain in general, it does correlate with pain in patients with HIV-associated polyneuropathy.²²⁵ Additionally, a higher proportion of patients with fibromyalgia exhibit decreased ENFD findings consistent with small-fiber polyneuropathy.⁴⁵

Imaging

While imaging can provide confirmation of pathology that could contribute to pain, abnormalities noted on imaging may not correlate to pain intensity or explain the patient's pain complaints. Regardless, imaging could be used to rule out pathology or determine if another process is contributing to the pain state. In patients with chronic pain, research suggests that changes in brain structure and function can occur.⁶⁶ Radiolabeled ligands can identify altered function in specific neurotransmitter systems, and

magnetic resonance spectroscopy can reveal neurochemical changes in specific brain regions.^{66,184} However, due to high cost and limitations of these studies, brain imaging in chronic pain patients is not yet part of the routine workup.

Pharmacogenetic Testing

Many patients with chronic pain fail to respond to certain medications. Identifying an individual's genetic makeup, enzyme function, and other factors can help to tailor an optimal treatment plan based on the patient's ability to process and respond to medications.¹¹³ While these approaches are promising, testing is not yet widely available.^{113,175}

Outcome Measurement

Due to the inherent difficulty in assessing chronic pain and determining effectiveness of treatments, the Initiative on Methods, Measurement, and Pain Assessment in Clinical Trials (IMMPACT) was developed to provide guidelines on interpreting treatment outcomes.⁸⁰ The IMMPACT guideline, published by the American Pain Society, established four pain domains that should be taken into consideration in the assessment of pain and determination of treatment outcomes. These domains include (1) pain intensity (0 to 10 numeric rating scale), (2) physical functioning (multidimensional pain inventory and brief pain inventory interference scales), (3) emotional functioning (Beck depression inventory and profile of mood states), and (4) self-ratings of overall improvement (patient global impression of change scale). The panel recommends two or more methods be used to evaluate for improvement or worsening of pain.

Table 38.4 summarizes the various tools used in assessment of chronic pain.

TABLE 38.4 Pain Assessment Tools

Evaluation Domain	Instrument	Alternate Name	Evaluation	Description	Result/Score
Pain Intensity	Visual Analog Scale	VAS	Subjective pain intensity	Patient selects point on horizontal line (usually 100 mm) with left end correlating to no pain and right end correlating to worst pain imaginable	Score determined by measuring in mm from the left end of the line to the marked point
	Numeric Rating Scale/Verbal Rating Scale	NRS/ VRS	Pain intensity	Patient is asked to report current or past pain experience based on a numeric scale from 0 (no pain) to 10 (worst pain)	Number indicates pain score
	Wong-Baker Faces Scale	Faces Scale	Pain intensity in children or those with verbal language limitation	Series of six faces ranging from happy face at 0 (no pain) and crying face at 10 (worst pain)	Number correlates to facial expression/pain observed by evaluator
Physical Functioning/ Symptom Description	The Brief Pain Inventory and Brief Pain Inventory Short Form	BPI/ BPI-SF	Pain intensity and interference with activities	Assesses pain at lowest, highest, and average intensities; measures pain interference with seven daily activities (general, walking, work, mood, life enjoyment, relationship, and sleep)	Patient rates each category: 0 (no pain/interference) to 10 (excruciating pain/complete interference)

Continued

TABLE 38.4 Pain Assessment Tools—cont'd

Evaluation Domain	Instrument	Alternate Name	Evaluation	Description	Result/Score
Impact of Pain	West Haven–Yale Multidimensional Pain Inventory	WHY MPI, MPI	Impact of pain on life, social, and general activities	Three sections: Pain severity/interference Spouse's/significant other's response to patient's pain behavior Self-reported engagement in everyday activities	Scored from 0 (never) to 6 (always): Higher scores in section 1 and 2 suggest more severe pain and significant interference. Lower scores in daily activities suggest reduced engagement
	Short Form McGill Pain Questionnaire-2	SF-MPQ-2	Pain quality, intensity, and perception	Two scales: Sensory subscale: 11 descriptive words (temporal, spatial, pressure, etc.) Affective subscale: 4 words (tension, fear, etc.)	Each word description is rated from 0 (none) to 3 (severe). The sum of all scores reflects pain intensity
	Oswestry Disability Index	ODI	Functional level and disability in spine disorders and low back pain	10 sections: pain intensity, personal care, lifting, walking, sitting, standing, sleeping, social life, sex life, traveling; each scored: 0 (no pain /issues) to 5 (most severe pain/disability)	Total score is the sum of all scores, multiplied by 2 to obtain percentage; higher score indicates more disability
	Pain Disability Index	PDI	Chronic pain interference with life activities	Seven areas of life affected: family/home responsibilities, recreation, social activity, occupation, sexual behavior, self-care, basic life activities	Scored from 1 (no disability) to 10 (highest disability) in each area; higher score indicates greater disability
Pain Impact With Specific Condition	Roland-Morris Disability Questionnaire	RMDQ	Physical disability in low back pain	24 statements related to activities impacted by back pain (walking, dressing, rising, etc.); responders select the statement that applies to them	Total score ranges from 0 (no disability) to 24 (maximum disability)
	Chronic Low Back Pain Impact Questionnaire	CLBP-IQ	Impact of pain in patients with low back pain	Assess five domains (physical function, emotional well-being, social function, sleep disturbances, cognitive function) each with its own set of questions	Each question is scored from 0 to 4; higher total scores indicate worse impact of low back pain on responder's life
Emotional Distress	Beck Depression and Anxiety Inventory	BDI	Depression and anxiety	21 self-reported questions each, covering physical symptoms (dizziness) and emotional symptoms (fear of death)	Each question is rated from 0 to 3; higher total score indicates higher severity of depression or anxiety
	Hospital Anxiety and Depression Scale	HADS	Anxiety and depression	Seven questions for anxiety and seven questions for depression; each section is scored separately	Each section is scored from 0 (not at all) to 3 (all the time); condition rated by scores: 8 to 10: mild, 11 to 14: moderate, 15 to 21: severe
General Health	Short Form–36 Health Survey	SF-36	General health status	Eight sections measuring physical, emotional, social, and mental health functioning along with general health perception, body pain, and vitality	Each item within the subscale is scored from 0 to 100, with a higher score denoting better health or functioning
Change in Condition	Global Rating of Change	GROC	Patient-reported outcome Rating of changes in health over time	One question used to score self-perceived improvement from start of treatment	Scored on a 15-point numeric system from -7 (worst) to +7 (best)

Management

Pain Treatment Services

There are several types of pain programs identified by the IASP. These include multidisciplinary pain centers, multidisciplinary pain clinics, and single provider practice. The single provider practice is managed by a provider who has completed specialty medicine training or is certified in pain management by the appropriate organization. Patient evaluation and assessment are limited based on the clinicians' medical training, so they should refer patients to a multidisciplinary pain center for issues that exceed their capability. In a multidisciplinary pain clinic, team members from various specialties independently assess and treat patients, sharing information as needed; conversely, providers in a multidisciplinary center collaborate on a comprehensive evaluation and treatment options. This evaluation includes a medical assessment, psychosocial evaluation, and vocational evaluation (if needed). Thus the ideal treatment program for patients with complicated chronic pain is a multidisciplinary pain center.¹⁹² Table 38.5 outlines recommended team members of a multidisciplinary pain center.⁵⁸

Medications

NSAIDs and Acetaminophen

Nonsteroidal antiinflammatory drugs (NSAIDs) and acetaminophen, also known as paracetamol, are the most commonly used

analgesic drugs worldwide.¹¹⁷ This trend is partly influenced by the framework established by the World Health Organization (WHO) step ladder approach for analgesic treatment, with acetaminophen and NSAIDs as the first-line option for mild pain. This guideline was first established in 1986 for the management of cancer pain and later extended to nonmalignant pain. Following the opioid crisis in the 1990s, many clinicians steered away from opioid use for chronic pain, unfortunately leaving NSAIDs and acetaminophen as the primary choice for all levels of pain.

NSAIDs inhibit cyclooxygenase (COX) and consequently inflammation, thereby reducing pain associated with inflammation. The COX-1 enzyme is important in gastric protection, kidney function, and vascular homeostasis; the COX-2 enzyme is involved in the inflammatory cascade. Because NSAIDs inhibit both COX-1 and COX-2 enzymes, they can compromise kidney function, cause gastrointestinal bleeding, increase the risk of atrial fibrillation, and interfere with the cardioprotective effect of aspirin. For these reasons, NSAIDs are not recommended for long-term use, especially in the geriatric population.

Newer generation NSAIDs such as celecoxib, valdecoxib, and rofecoxib (the latter two were later taken off the market) selectively inhibit COX-2 and thereby have fewer gastroduodenal and renal side effects. Additionally, they were reported to have little to no effect on platelet aggregation or bleeding time commonly found with traditional NSAIDs. However, selective COX-2 inhibitors still carry some cardiac, renal, and gastroduodenal risks and are not entirely safe for chronic use. Rofecoxib and valdecoxib

TABLE 38.5 Interdisciplinary Team Members^{58,126}

Pain physicians (physiatrist, neurologist, anesthesiologist, psychiatrist, family physician, emergency physician, primary care provider)	Provide multifaceted approach to medical evaluation: performing a comprehensive medical exam, offering interventions when indicated, and formulating the treatment plan tailored to the patient's needs; these physicians also coordinate with other members of the treatment team
Pain/addiction psychiatrist	May assist with cooccurring mental health condition management (e.g., antidepressant titration) and assessment for substance use disorder(s); may offer/titrate buprenorphine products when appropriate
Pain pharmacist	May coordinate patient progress and care with the team, especially when starting and titrating medications
Pain psychologist	Assessment and therapeutic interventions focus on both cognitive and behavioral factors related to pain; psychological intervention focuses on unlearning maladaptive responses and reactions to pain, while fostering self-efficacy, wellness, improved coping, perceived control, decreased catastrophizing, and acceptance
Physical therapist	Specialize in gait training, locomotion, core stability, lower extremity biomechanics, and functional mobility, as well as instrumental activities of daily living (IADLs) and traditional focus on ADLs (i.e., bed mobility and transfers); they are experts in the development of aerobic conditioning programs aimed at improving cardiopulmonary health and endurance
Occupational therapist	Typically concentrate on educating patients regarding proper posture and ergonomics related to upper limb functional activities such as lifting and computer use; they address upper extremity-related ADLs including feeding, hygiene, grooming, bathing, and dressing
Nurse facilitator	The program nurse or nursing staff member plays an essential role in coordinating patient progress and care during evaluation, treatment, and ongoing follow-up evaluation; nursing plays a crucial role in educational aspects of treatment, including basic instruction on pain pathways, pharmacology, nutrition, and sleep hygiene
Social worker	Social workers are integral to the team to assist with housing, transportation, financial aid, resources, and coordinate with team as these elements relate to function, prognosis, and quality of life
Vocational rehabilitation	Vocational counselors participate in the analysis of current or previous job descriptions, provide suggestions for work accommodation or modification, and, if necessary, facilitate vocational testing and targeted retraining; at the end of the rehabilitation process, vocational rehabilitation can help to coordinate functional capacity evaluation testing and finalize return to work issues (i.e., restrictions and level of work)

were taken off the market due to cardiac toxicities. Meloxicam and etodolac are unique COX-2 inhibitors in that at lower dosages (i.e., meloxicam 7.5 mg daily) they are more selective at the COX-2 site but have greater effect at COX-1 receptors at higher doses. Diclofenac also has relative selectivity of COX-2 inhibition and is available in both oral and topical formulations. If NSAIDs are deemed necessary for the occasional exacerbation of chronic pain, judicious monitoring for efficacy and side effects is recommended.

Acetaminophen is recommended by the American Geriatrics Society (AGS) as the drug of choice for mild to moderate musculoskeletal pain and osteoarthritis.²¹⁵ The analgesic properties of acetaminophen are similar to those of NSAIDs, but acetaminophen does not exhibit antiinflammatory effects in the same way as NSAIDs. Acetaminophen has both peripheral (COX) and central (COX, serotonergic descending neuronal pathway, L-arginine–nitric oxide pathway, cannabinoid system) mechanisms.¹⁵³ Acetaminophen has fewer side effects compared to NSAIDs when taken as prescribed but may cause liver necrosis if taken inappropriately (>4 g/day or 125 mg/kg body mass).²⁰³

Muscle Relaxants

Muscle relaxants, also referred to as antispasmodics or antispastics, are commonly used to relieve pain and stiffness from musculoskeletal injuries such as direct muscle trauma, strain from poor posture, overexertion, and abnormal gait. Most skeletal muscle relaxants, except dantrolene sodium, act on the central nervous system to produce sedation and consequently systemic relaxation of all the muscles in the body, not just the injured or affected muscle. Cyclobenzaprine is structurally similar to tricyclic antidepressants and shares anticholinergic side effects of drowsiness, dizziness, and dry mouth. Dantrolene prevents calcium release by the sarcoplasmic reticulum and is used to treat spasticity caused by spinal cord injury, stroke, and multiple sclerosis. Of note, dantrolene can cause liver damage at higher doses. Diazepam and carisoprodol have been associated with dependence and abuse, making them less useful for chronic treatment.²⁰⁸ There is inadequate evidence for the effectiveness of the long-term use of muscle relaxants.⁴⁹

Benzodiazepines

Benzodiazepines act at the GABA-A receptors in the central nervous system. They are primarily used in the management of anxiety disorders, alcohol withdrawal, insomnia, spasticity, and some seizure disorders. Some studies suggest they exhibit analgesic effects in burning mouth syndrome,²²⁶ hypertonia associated with stiff person syndrome,³¹ and phantom limb syndrome.²² Studies evaluating the use of benzodiazepines for other pain complaints such as fibromyalgia, low back pain, abdominal pain, and headaches were inconclusive and/or lack efficacy and safety. Benzodiazepines have a significant side effect profile, including sedation, cognitive dysfunction, dependence, abuse, and high mortality from accidental or intentional overdose. Therefore their use in chronic pain management is typically discouraged.²⁶⁴

Adjuvants

Both the WHO and the IASP endorse the use of adjuvants, along with conventional medication, in the management of pain.¹⁰⁰ Adjuvant analgesics, also known as coanalgesics, are medications that were not primarily designed for pain control but can be used effectively for this purpose. Examples of adjuvant drugs include

antidepressants and anticonvulsants. Adjuvants act on the excitatory (e.g., substance P, glutamate), inhibitory (e.g., GABA), or other neurotransmitters that modulate the pain experience (e.g., serotonin, norepinephrine). They are particularly useful for neuropathic pain, cancer pain, and central pain syndromes such as fibromyalgia. These medications can complement other pain treatments or be used independently. The dose of adjuvants needed to achieve analgesia is often lower than the dose prescribed for their original indications. Therefore the side effects from adjuvants are generally minimal and avoid the risks opioids carry, including tolerance and dependence. A more detailed list of various adjuvants used in the management of chronic pain is listed in Table 38.6.

Antiarrhythmics/Anesthetics

Mexiletine, an antiarrhythmic oral lidocaine analog, acts as a sodium channel blocker in peripheral nerves and has been used in a variety of chronic neuropathic pain conditions as well as fibromyalgia.²¹⁴ There is conflicting evidence that mexiletine reduces accompanying paresthesia or dysesthesia and ameliorates sleep disturbances associated with painful diabetic neuropathy.^{47,182,240,256} The most common side effects are gastrointestinal symptoms (nausea, vomiting, gastritis) and central nervous system effects (dizziness, lightheadedness, nervousness, changes in sleep habits), but these were usually mild and tolerable. Mexiletine has serious cardiac side effects at higher doses, including exacerbation of arrhythmias and bradycardia, so it should be avoided in patients with a history of cardiac disease.²¹⁴

Ketamine is an anesthetic agent with NMDA receptor antagonistic properties, but it also acts on mu-opioid, cholinergic, nicotinic, dopaminergic, serotonergic, and voltage-gated calcium channels receptors.⁶² Common side effects include hallucinations, delirium, nausea, and vomiting. Ketamine is given intravenously (IV) as an infusion at a subanesthetic dose and is an emerging treatment for chronic neuropathic pain and neuropathic cancer pain.⁶² Although there is evidence to support the use of ketamine for chronic pain, the level of evidence varies by condition and dosing.⁵⁷ A lack of standardized dosing is a significant barrier to effective research of this treatment. Ketamine has been studied in complex regional pain syndrome,⁵⁹ but only low to moderate quality of evidence was found in systematic reviews to support the use to treat this condition.²⁶⁵ There are studies suggesting that IV ketamine can counteract opioid-induced hyperalgesia to reduced total opioid dose required.¹³⁹ Oral ketamine, also shown to counteract opioid-induced hyperalgesia, is not recommended due to risk of toxicities, misuse, abuse, and diversion. Studies on intranasal ketamine for cancer pain are promising but limited.⁴⁶

Calcium Channel Blockers

Calcium channel blockers may reduce pain signal transduction and have been used adjunctively to opioid therapy. Verapamil has been shown to be beneficial as a prophylactic treatment for migraine and cluster headaches and in the management of chronic pain.^{200,234} The mechanism of action is postulated to involve inhibition of the release of pronociceptive mediators (substance P, CGRP, glutamate). Ziconotide is a synthetic version of a conopeptide found in the venom of the marine snail. It blocks the neuronal N-type voltage-sensitive calcium channels and has been shown to provide significant relief to patients with chronic pain refractory to other therapies.¹⁷³ Intrathecal ziconotide was approved by the US Food and Drug Administration (FDA)

TABLE 38.6 Adjuvant Medications Commonly Used in Chronic Pain

Classification	Drug	Dosing (Parenteral route unless noted)	Uses	Comments
Anticonvulsants				
Gabapentinoid	Gabapentin	300 mg tid	Neuropathic pain, chronic headache	Most commonly used adjuvant for neuropathic pain. ¹³³ Start 300 mg/day. Titrate up as needed to max recommended dose 1800 mg/day. Gradually taper off over at least 1 week if ineffective.
	Pregabalin	150 mg/day (divided in bid or tid dosing)	Postherpetic neuralgia, diabetic neuropathy, fibromyalgia	FDA approved for diabetic neuropathy and postherpetic neuralgia. Can titrate to 300 mg/day, max 600 mg/day. Better side effect profile than gabapentin. Gradually taper off over at least 1 week ineffective.
Fatty acid derivative	Valproic acid	250 mg tid with target of 1000–2500 mg daily	Neuropathic pain, fibromyalgia	Insufficient evidence to support use as first-line preventive treatment ¹⁰² ; limited by gradual titration.
Carbonic anhydrase inhibitor	Topiramate	25 mg/day titrate to 50 mg bid	Headache, neuropathic pain	Migraine prophylaxis. Conflicting evidence in its usefulness for neuropathic pain. ^{187,260}
Dibenzazepine	Carbamazepine	300–800 mg/day in 2 divided doses	Trigeminal neuralgia, neuropathic pain, fibromyalgia	FDA approved for neuropathic pain. First line for trigeminal neuralgia. ²⁷⁰ Probably effective for other types of neuropathic pain. ^{81,132}
Antidepressants				
SNRI	Duloxetine	40–60 mg (qd to bid, max 120 mg/day for pain)	Diabetic peripheral neuropathy, fibromyalgia, chronic low back pain, osteoarthritis pain	FDA approved for chronic musculoskeletal pain, osteoarthritis, low back pain, fibromyalgia, diabetic polyneuropathy. Effective centrally acting general analgesic. ²³² First-line option for chronic low back pain (analgesic dose 60 mg/day with initial dose at 20–30 mg bid). ^{33,118,232}
	Venlafaxine	37.5–75 mg bid to tid (max 225 mg/day)	Neuropathic pain	High lipophilicity; safe and well tolerated. ⁸
	Milnacipran	12.5 mg qd initial dose, titrate to 50 mg bid after 1 week	Fibromyalgia	FDA approved for treatment of fibromyalgia. ³³
SSRI	Sertraline	25 mg qd (max 200 mg/day)	Chronic pain	May have some effects on chronic pain conditions but studies are limited. ⁷⁴ There is no current reliable evidence of long-term efficacy. ³³ Monitor serotonin syndrome and QTc prolongation.
	Fluoxetine	10–20 mg qd		No significant analgia when used alone but may modulate tolerance and dependence when used concurrently with opioids. ¹⁰⁸
	Citalopram	40 mg/day	Chronic pain	Inconclusive evidence for effective use in chronic pain condition ³³ ; less activating.
SARIs	Trazadone	50–100 mg qhs	Sleep disturbance	Helps regulate sleep-wake cycle; sedating effect.
Tricyclics	Amitriptyline Nortriptyline	10–25 mg qhs Max 150 mg/day	Migraines, fibromyalgia, neuropathic pain	May also help with sleep issues. ⁷⁴
Anxiolytics				
Benzodiazepines	Valium	2–10 mg qd-bid	Acute muscle spasm	Short term use. Risk of dependence, misuse. Avoid concurrent use with opioids. Taper dose to discontinue. ^{49,199} Typically not preferred for chronic pain.
	Clonazepam	0.5–2 mg qhs	Neuropathic pain, myoclonus, phantom limb pain, restless leg syndrome	Some efficacy with management of neuropathic pain and phantom limb pain. May be helpful in restless leg syndrome and burning mouth syndrome. Risk of dependence, misuse. Avoid concurrent use with opioids. Taper dose to discontinue. ²²

Continued

TABLE 38.6 Adjuvant Medications Commonly Used in Chronic Pain—cont'd

Classification	Drug	Dosing (Parenteral route unless noted)	Uses	Comments
Antipsychotics				
Atypical antipsychotics	Quetiapine	25–200 mg/day	Migraine headache	Sedating effect. Limited evidence for treatment of pain. May be helpful in improving sleep. Further studies needed. ²²³
	Olanzapine	2.5–10 mg/day	Fibromyalgia, migraine headache	May be helpful for those with central sensitization or psychiatric comorbidities. May be helpful in improving sleep. Additional prospective trials are needed. ²²³
Antispasmodics				
Sarcoplasmic reticulum Ca ⁺⁺ inhibitor	Dantrolene	25–100 mg qd-tid (max 400 mg/day)	Spasticity	Used in pain associated with spasticity. Side effects of severe hepatotoxicity. ^{49,199}
GABA _B agonist	Baclofen	5 mg tid (max 80 mg/day)	Spasticity, muscle spasm	Preventive, can cause sedation and lower seizure threshold. ^{49,199}
Alpha ₂ agonist	Tizanidine	2–6 mg tid to qid (max 36 mg/day)	Spasticity, Fibromyalgia	Helpful in withdrawal syndrome. May have negative cognitive effect.
Norepinephrine potentiation and serotonin receptor agonist	Cyclobenzaprine	5–10 mg qd to tid	Fibromyalgia, myofascial pain, low back pain	May be helpful for short term (≤3 weeks) management of muscle spasm and musculoskeletal pain. Longer uses for myofascial pain and low back pain have shown some efficacy. ^{56,153,202}
Natural supplements	Melatonin	1 mg qhs (max 10mg)	Sleep disruption	Effect on circadian rhythm. May improve sleep related fatigue. ⁷
	Alpha-lipoic acid	600–1800 mg/day	Diabetic neuropathy	Antioxidant. Liquid formulation has greater bioavailability. ²¹⁷
Toxins	Botulinum toxin	Variable dosing	Spasticity, chronic migraines, ischemic pain, trigeminal neuralgia, pelvic floor pain	Good safety profile for migraine prevention compared to other treatment option. Promising data for treatment of other chronic pain conditions in which other treatment have failed. ^{115,116,148,210,213,230,236}

bid, Twice per day; *GABA*, gamma-aminobutyric acid; *qd*, once per day; *qhs*, at bedtime; *SARI*, serotonin receptor antagonists and reuptake inhibitors; *SNRI*, serotonin norepinephrine reuptake inhibitors; *SSRI*, selective serotonin reuptake inhibitors; *tid*, three times/day.

for the treatment of severe chronic pain and is used alone or in combination with other intrathecal analgesics.^{207,233}

Neurotoxin

Botulinum toxin (BTX) is a neurotoxin that blocks presynaptic release of acetylcholine, causing weakness or paralysis of skeletal muscles. It also has roles in the modulation of proinflammatory pain mediators. Experimentally, BTX suppresses other neurotransmitters, including norepinephrine, substance P, and CGRP.²³⁰ BTX type A has been FDA approved for the treatment of refractory chronic migraines, though there may be utility in treating trigeminal neuralgia and other neuropathic pain conditions.^{115,213,230}

Opioids

Opiates are substances that are developed from the opium poppy plant; these include morphine, codeine, and heroin. Opioids refer to both natural (opiates) and synthetic/semisynthetic substances. Synthetic opioids include fentanyl and methadone, while semi-synthetic substances include hydrocodone and oxycodone. These

all bind to the opioid receptors (mu, delta, kappa) belonging to a G-protein receptor family. Opioid receptors are found in the brain, spinal cord, and other organs. Opioids work by decreasing presynaptic calcium flow into the cell and inhibiting the release of excitatory neurotransmitters (serotonin, norepinephrine, substance P, and glutamate). Postsynaptically, they increase potassium efflux, resulting in hyperpolarization of the neuron and decreasing synaptic transmission. At the brainstem level, opioids inhibit GABAergic transmission, leading to excitation of descending inhibition.¹⁸³

Opioids are potent analgesics used for the treatment of moderate to severe pain (Table 38.7). However, long-term use of opioids may result in adrenal insufficiency, respiratory depression, sex hormone disruption, osteoporosis, opioid-induced bowel dysfunction, opioid-induced hyperalgesia, and immunosuppression. Other potential risks of long-term use include misuse, dependence, and opioid use disorder (OUD). Newer long-acting formulations of hydrocodone (Hysingla ER, Zohydro ER), hydromorphone (Exalgo ER), and oxycodone (Xtampza ER) have abuse-deterrent properties such as physical and/or

TABLE 38.7 Commonly Used Opioids^{78,79,89,248}

Opioid (mu-receptor agonist)	Dosage (mg) Equivalent to Morphine (30 mg PO)	Dosage IV (mg)	MME	Duration of Action (hr.)	Comments
Codeine	180–200		0.15	4	Weak opioid due to weak mu affinity. Metabolized by CYP2D6 to morphine. Used as single agent or combined with other medication.
Fentanyl transdermal (μg/hr)	12.5/9 (μg/hr)		2.4	2	Synthetic. Extensive metabolism in liver, mostly by CYP3A4. Transdermal patch has 6–12 hr. onset of action.
Hydrocodone Long acting (Zohydro ER, Hysingla ER)	30		1	3–5 12–24	Parent drug has weak mu affinity but converts by CYP2D6 to hydromorphone, which has strong mu affinity. Used as single agent or combined with other medication.
Hydromorphone (Dilaudid) Long acting (Exalgo ER)	4–6	1.5	5	3–4 12	Semisynthetic opioid with some activity at delta receptor. Glucuronidation metabolism. Extensively metabolized by liver.
Methadone (Dolophine, Methadose)	Variable		4.7	4–8	Synthetic. Also NMDA antagonist. Metabolized by liver by CYP3A4 (primary) and CYP2D6 (secondary). Analgesic effect is 4–8 hr. but plasma elimination is substantially longer at 8–59 hr. May cause ST prolongation.
Morphine Long acting (Avinza, Kadian, MS Contin)	30	10	1	3–4 8–12	Glucuronidation (predominant) and demethylation metabolism. May induce urticaria.
Oxycodone (Roxicodone) Long acting (Oxycontin, Xtampza ER)	15–20		1.5	3–5 12	Also acts at kappa receptor, glucuronidation, and CYP2D6 metabolism. Parent compound is primary analgesic (not a prodrug).
Oxymorphone (Opana) Long acting (Opana ER)	10	1	3	3–5 12	Low affinity with kappa and delta receptors, not affected by CYP2D6 or CYP3A4.
Tapentadol (Nucynta)	100		0.4	12	Atypical opioid with mu agonist and norepinephrine reuptake inhibition.
Tramadol (Ultram, ConZip, Qdolo)	150–300		0.2	4–6	Atypical opioid with partial mu agonist as well as central GABA, norepinephrine and serotonergic reuptake inhibition. Maximum 400 mg/day.

CYP2D6, Cytochrome P450 2D6 (enzyme that is mainly expressed in the liver but also highly expressed in areas of the central nervous system); *CYP3A4*, cytochrome P450 3A4 (enzyme that is mainly expressed in the liver and the intestine); *ER*, extended release; *GABA*, gamma-aminobutyric acid; *MME*, morphine milligram equivalent; *NMDA*, N-methyl-D-aspartate (central nervous system receptor that is a glutamate primary excitatory neurotransmitter).

NOTE: 1) All opioids listed act at the mu receptor. Some also act at the delta and kappa receptors.

2) Drugs that are metabolized by CYP2D6 are susceptible to drug-drug interactions to inhibitors (i.e., bupropion, celecoxib, cimetidine, cocaine) and inducers (i.e., dexamethasone, rifampin).

chemical barriers to prevent chewing, crushing, grinding, or dissolving tablets. In addition, an opioid antagonist can be added to the parent drug (e.g., Suboxone [buprenorphine and naltrexone]) to deter misuse. Since the added opioid antagonist (naloxone) is relatively inactive when taken as prescribed (sublingually), it does not interfere with the parent drug. However, when the combination drug is manipulated for purposes of injection or inhalation, the naloxone becomes active and counteracts the action of the parent agonist drug.⁴⁴ A list of common opioids used in chronic pain is presented in Table 38.7.

Guidelines for opioid prescription. In 2016, in response to the opioid epidemic, the Centers for Disease Control and Prevention (CDC) published practice guidelines for opioid use for chronic nonmalignant pain to help primary care clinicians weigh the risks and benefits of prescribing opioids in the outpatient

setting. These recommendations would not apply to sickle cell disease, cancer pain, palliative care, and end-of-life-care. CDC guidelines include 12 recommendations highlighting a patient-centered approach for initiation of opioids with guidance for managing and tapering patients on chronic opioid therapy. Additional guidance on appropriate application of the guidelines and tapering for patients no longer benefiting or when risks outweigh benefits to continue opioid therapy were published in 2019 by the CDC and Health and Human Services, respectively.⁷⁹ In 2022, the updated guidelines expanded to acute and subacute pain (Table 38.8). The 2022 guideline aims to promote equitable access to effective, informed, individualized, and safe pain management that improves patients' function and quality of life, while clarifying and reducing the risks associated with opioid use.⁷⁸ These principles are outlined in Table 38.9.

TABLE 38.8

Centers for Disease Control and Prevention Clinical Practice Guideline for Prescribing Opioids for Pain^{78,79}

Recommendations for prescribing opioids for outpatients with pain, excluding pain management related to sickle cell disease, cancer-related pain treatment, palliative care, and end-of-life care

- 1 Nonopioid therapies are at least as effective as opioids for many common types of acute pain.
- 2 Nonopioid therapies are preferred for subacute and chronic pain.
- 3 When starting opioid therapy for acute, subacute, or chronic pain, clinicians should prescribe immediate-release opioids instead of extended-release and long-acting opioids.
- 4 When opioids are initiated for opioid-naïve patients with acute, subacute, or chronic pain, clinicians should prescribe the lowest effective dosage.
- 5 For patients already receiving opioid therapy, clinicians should carefully weigh benefits and risks and exercise care when changing opioid dosage. Unless there are indications of a life-threatening issue such as warning signs of impending overdose (e.g., confusion, sedation, or slurred speech), opioid therapy should not be discontinued abruptly, and clinicians should not rapidly reduce opioid dosages from higher dosages.
- 6 When opioids are needed for acute pain, clinicians should prescribe no greater quantity than needed for the expected duration of pain severe enough to require opioids.
- 7 Clinicians should evaluate benefits and risks with patients within 1–4 weeks of starting opioid therapy for subacute or chronic pain or of dosage escalation and regularly reevaluate benefits and risks of continued opioid therapy.
- 8 Before starting and periodically during continuation of opioid therapy, clinicians should evaluate risk for opioid-related harms and discuss risk with patients. Clinicians should work with patients to incorporate into the management plan strategies to mitigate risk, including offering naloxone.
- 9 When prescribing initial opioid therapy for acute, subacute, or chronic pain, and periodically during opioid therapy for chronic pain, clinicians should review the patient's history of controlled substance prescriptions using state prescription drug monitoring program data to determine whether the patient is receiving opioid dosages or combinations that put the patient at high risk for overdose.
- 10 When prescribing opioids for subacute or chronic pain, clinicians should consider the benefits and risks of toxicology testing to assess for prescribed medications as well as other prescribed and nonprescribed controlled substances.
- 11 Clinicians should use particular caution when prescribing opioid pain medication and benzodiazepines concurrently and consider whether benefits outweigh risks of concurrent prescribing of opioids and other central nervous system depressants.
- 12 Clinicians should offer or arrange treatment with evidence-based medications to treat patients with opioid use disorder. Detoxification on its own, without medications for opioid use disorder, is not recommended for opioid use disorder because of increased risks for resuming drug use, overdose, and overdose death.

TABLE 38.9

Guiding Principles for Implementation of the Centers for Disease Control and Prevention (CDC) Clinical Practice Guideline for Prescribing Opioids for Pain (Implemented in 2022)^{78,79}

- 1 Acute, subacute, and chronic pain needs to be appropriately assessed and treated independent of whether opioids are part of a treatment regimen.
- 2 Recommendations are voluntary and are intended to support, not supplant, individualized, person-centered care. Flexibility to meet the care needs and the clinical circumstances of a specific patient is paramount.
- 3 A multimodal and multidisciplinary approach to pain management attending to the physical health, behavioral health, long-term services and supports, and expected health outcomes and well-being of each person is critical.
- 4 Special attention should be given to avoid misapplying this clinical practice guideline beyond its intended use or implementing policies purportedly derived from it that might lead to unintended and potentially harmful consequences for patients.
- 5 Clinicians, practices, health systems, and payers should vigilantly attend to health inequities; provide culturally and linguistically appropriate communication, including communication that is accessible to persons with disabilities; and ensure access to an appropriate, affordable, diversified, coordinated, and effective nonpharmacologic and pharmacologic pain management regimen for all persons.

As a response to the opioid crisis, there have been efforts to curtail the use of opioids for chronic pain. Some studies suggest that opioids are not superior to NSAIDs in the treatment of pain.^{145,220} However, some of these studies, including the Strategies for Prescribing Analgesics Comparative Effectiveness trial, have the limitation in that the dose of opioids given was low, the population studied had mild to moderate pain and may not have

had an indication for opioids, and participants in the nonopioid arm were also concurrently treated with antidepressants and tramadol, a weak opioid agonist.⁹¹

Opioid therapy, opioid use disorder, and buprenorphine considerations. Long-term opioid therapy (LTOT) for chronic cancer and noncancer pain may be appropriate and useful for some. For others, LTOT may be suboptimal in providing adequate

pain control. There is very low-quality evidence of the long-term efficacy, tolerability, and safety of opioids for chronic low back, osteoarthritis, and diabetic polyneuropathic pain. Research has shown one-third of patients benefit from LTOT.³² Strong support for opioid overutilization comes from the pain treatment in chronic nonmalignant pain settings. Higher opioid doses in this population are generally associated with mental health conditions (anxiety or depression) along with substance use disorders.¹⁷¹ Emerging data suggest that the prolonged use of opioids leads to hypogonadism, fractures, and cognitive blunting.²⁴¹ Provocative data from the laboratory also suggest opioids may accelerate tumor growth.⁶ Finally, some studies have shown that opioid use does not improve function or quality of life in some long-term cancer survivors or those with chronic nonmalignant pain.⁴¹

Opioid tapering and/or discontinuation among many patients on LTOT often leads to even worse pain control and function, leaving the patients and providers with few treatment choices. Complex persistent opioid dependence (CPOD) was recently offered as a heuristic to explain this clinical challenge exemplified by the ineffectiveness of both LTOT and opioid tapering. Additionally, in assessing patients for OUD, it is important to note that 2 of the 11 criteria, withdrawal and tolerance, are not applied when assessing those who use opioids only under medical supervision (Table 38.10). Meanwhile, some researchers have argued that this adjustment is inadequate when addressing the realities of health care professionals diagnosing OUD among pain patients because behaviors and attitudes associated with OUD may also be associated with reasons other than abuse, such as the need for pain relief. Gorfinkel et al. point out that DSM-5 criteria such as “opioids are often taken in a larger amount or over a longer period of time than intended” and “craving or a strong desire to use opioids” may be associated with escalating or unabating pain. Other criteria may be similarly ambiguous when applied to pain patients, especially if there is not additional probing as to the reasons for their answer.¹⁰⁵

Obtaining a detailed history and continuing to assess patients on subsequent visits is vital in diagnosing appropriately. The sensitivity analysis demonstrated that the DSM-5 criteria related to tolerance and withdrawal will have an impact on the observed rate of OUD among patients without nonmedical opioid use. The rate of moderate or severe OUD among patients with no nonmedical opioid use increases from 6% when tolerance and withdrawal criteria are not applied to 11.5% when tolerance and withdrawal criteria are inadvisably applied, an increase of 92%. This variation in rates underscores the importance of consistently defining criteria for the assessment of OUD, especially in the context of patients prescribed opioids for chronic pain.³⁷ CPOD may be a more appropriate diagnosis for those experiencing tolerance/withdrawal while taking opioids as prescribed, under medical supervision. Opioid-induced chronic pain syndrome (OICP) is offered as an alternate clinical diagnostic term to CPOD because CPOD may have poor patient acceptance due to stigma toward addiction and clinical confounding with OUD, a related but separate clinical entity. OICP with LTOT is conceptualized as a recoverable iatrogenic problem that can be managed by pain providers.¹⁶²

In such cases, buprenorphine products may be considered to help transition from full opioid agonist medication(s). As a partial opioid agonist, buprenorphine has a ceiling effect with less risk of overdose. Buprenorphine may be utilized for pain management in cases where use of a full opioid agonist is suboptimal (e.g., patient is using cannabis, prescribed other depressant medications, drinking alcohol). Belbuca and Butrans are FDA approved for chronic

pain, while Suboxone, Subutex, and Sublocade are FDA approved for OUD. Table 38.11 lists various formulations of buprenorphine. These products can be used interchangeably/off label for pain when titrating (e.g., utilizing Belbuca to transition to Subutex). Table 38.12 outlines a possible titration schedule for buprenorphine (low-dose initiation). This may be started while the patient is still taking full-agonist opioids medications.

Opioid antagonist. Naltrexone is a nonselective competitive antagonist with high affinity for mu-opioid receptors. The use of naltrexone in the treatment of chronic pain has recently gained much attention. Low-dose naltrexone (1.5–4.5 mg/day) can reduce the glial inflammatory response by modulating Toll-like receptor 4 signaling as well as systemically upregulating endogenous opioid signaling by blocking transient opioid receptors.²⁴⁶ Low-dose naltrexone (up to 4.5 mg/day) has been shown to be effective in the symptomatic management of fibromyalgia²⁶⁷ as well as other chronic pain conditions such as multiple sclerosis and Crohn disease.¹⁹⁸ Ultralow-dose naltrexone (1 µg/day) acts on filamin A, a scaffolding protein involved in mu-opioid receptor signaling and has been used to reduce the total requirement of opioids postoperatively by potentiating the analgesic effects of opioids as well as reducing opioid-related side effects.²⁴⁶ Overall, studies on the use of low-dose naltrexone in the treatment of pain are promising but still limited.¹¹² Of note, naltrexone at higher doses (e.g., 50 mg oral and 380 mg injectable naltrexone [Vivitrol]) is used for alcohol disorders and OUD.

NMDA Antagonist

Memantine is an NMDA antagonist that has been approved for the treatment of Alzheimer disease but can be helpful in pain management. Among the NMDA antagonists that have been used for neuropathic pain, memantine has the most favorable side effect profile.²⁰¹ A small number of studies have shown that memantine has been effective in the management of fibromyalgia,¹⁹⁰ postoperative neuralgia, and phantom limb pain,¹⁸⁸ but these studies were small and subjected to potential biases.¹⁴⁹ Larger-scale studies are needed to explore the potential of this drug in the management of chronic pain.

Topicals

Topical medications are safer options compared to oral medications. Research suggests that locally applied topical agents can achieve therapeutic concentrations in specific tissues while maintaining low levels in the bloodstream. This approach may help reduce the risk of systemic toxicity.²³⁷

Topical NSAIDs are becoming more popular in the United States. Topical NSAIDs such as diclofenac and ketoprofen patch/gel work by inhibiting prostaglandin synthesis, affecting the lipoxigenase pathway, modulating excitatory amino acids, and influencing G protein-mediated signal transduction. Studies have shown that topical NSAIDs can provide good pain relief in patients with osteoarthritis and other musculoskeletal conditions, but evidence of their effectiveness in other pain conditions is lacking.⁷¹

Topical nitroglycerin (NTG), typically used for the prevention of angina, has been shown to be effective for the localized treatment of musculoskeletal pain. NTG may be converted to nitric oxide (NO), and the role of NO in pain and tissue healing is unclear. However, experiments have shown that NO enhances new tissue synthesis through increasing local blood flow, host defense, and collagen synthesis.³⁶ Nitroglycerin ointment has been used for pain-related chronic tendinopathies with good efficacy. In addition to relieving pain during activities of daily living, topical NTG also

TABLE
38.10Opioid Use Disorder Criteria¹¹

DSM-5 Criteria for Diagnosis of Opioid Use Disorder

Diagnostic Criteria*

These criteria not considered to be met for those individuals taking opioids solely under appropriate medical supervision.

Check all that apply

	Opioids are often taken in larger amounts or over a longer period of time than intended.
	There is a persistent desire or unsuccessful efforts to cut down or control opioid use.
	A great deal of time is spent in activities necessary to obtain the opioid, use the opioid, or recover from its effects.
	Craving, or a strong desire to use opioids.
	Recurrent opioid use resulting in failure to fulfill major role obligations at work, school or home.
	Continued opioid use despite having persistent or recurrent social or interpersonal problems caused or exacerbated by the effects of opioids.
	Important social, occupational or recreational activities are given up or reduced because of opioid use.
	Recurrent opioid use in situations in which it is physically hazardous
	Continued use despite knowledge of having a persistent or recurrent physical or psychological problem that is likely to have been caused or exacerbated by opioids.
	*Tolerance, as defined by either of the following: (a) a need for markedly increased amounts of opioids to achieve intoxication or desired effect (b) markedly diminished effect with continued use of the same amount of an opioid
	*Withdrawal, as manifested by either of the following: (a) the characteristic opioid withdrawal syndrome (b) the same (or a closely related) substance are taken to relieve or avoid withdrawal symptoms

Total Number Boxes Checked: _____

Severity: **Mild:** 2–3 symptoms. **Moderate:** 4–5 symptoms. **Severe:** 6 or more symptoms

*Criteria from American Psychiatric Association (2013). Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition., Washington, DC, American Psychiatric Association page 541. For use outside of IT MATTRs Colorado, please contact ITMATTRsColorado@ucdenver.edu

Courtesy IT MATTRs Colorado.

TABLE 38.11 Various Formulations of Buprenorphine and Their Uses**Buprenorphine formulations FDA approved for chronic pain**

Belbuca	Buprenorphine buccal film (available doses: 75–900 µg bid)
Butrans	Buprenorphine transdermal patch (available doses: 5, 7.5, 10, 15, 20 µg/hr)
Norspan	Buprenorphine transdermal patch (available doses: 5, 10, 20, 25, 30, 40 µg/hr)
Buprenex	Buprenorphine IV injection

Belbuca is not used off label for opioid use disorder, though it may be used in low-dose initiations while hospitalized (e.g., see low-dose initiation template)

Buprenorphine formulations FDA approved for Opioid use disorder

Bunavail	Buprenorphine/naloxone buccal film
Subutex	Buprenorphine sublingual tablet
Suboxone	Buprenorphine/naloxone sublingual tablet or film
Zubsolv	Buprenorphine/naloxone sublingual tablet
Sublocade	Buprenorphine extended-release subcutaneous injection
Brixadi	Buprenorphine extended-release subcutaneous injection

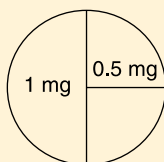
These products may be considered for off-label use for pain management

FDA, US Food and Drug Administration; IV, intravenous.

TABLE 38.12 Buprenorphine Low-Dose Initiation**Low-dose “micro-induction” buprenorphine start process**

Day	1	2	3	4	5	6	7
Buprenorphine dose	One quarter of a 2 mg tablet (0.5 mg) once	One quarter of a 2 mg tablet (0.5 mg) in AM and PM	One half of a 2 mg tablet (1 mg) in AM and PM	One full 2 mg tablet (2 mg) in AM and PM	Two full 2 mg tablets (4 mg) in AM and PM	Two full 2 mg tablets (4 mg) in AM, at noon, and in PM	FOUR full 2 mg tablets (8 mg) in AM and PM
Total daily buprenorphine dose	0.5 mg	1 mg	2 mg	4 mg	8 mg	12 mg	16 mg
Other opioid use	*Belbuca 300 mcg BID; 300 mcg TID; 600 mcg BID → 2 mg*				STOP		

How to cut 2 mg buprenorphine tablet for different dose amounts:



A low-dose “micro-induction” buprenorphine start allows you to begin buprenorphine while still taking another opioid, without experiencing any period of opioid withdrawal. It uses small doses of buprenorphine in the first three days to gradually nudge the other opioid off of the opioid receptors without precipitating withdrawal. This schedule is a good starting approach, but it is okay to speed up or slow-down this process if needed in collaboration with your medical provider.

The prescription comments will state: day 1 take one quarter tablet once, day 2 take one quarter tablet twice daily, day 3 take one half tablet twice daily, day 4 take one full tablet twice daily, day 5 take two full tablets twice daily, day 6 take two full tablets three times daily, day 7 take four full tablets twice daily

increases tendon strength.^{29,196} Studies for use of NTG in other pain conditions are limited, and additional long-term studies are needed to explore the potential use of topical NTG.

The topical formulation of lidocaine (5% patch), an amino-amide anesthetic, is FDA approved for postherpetic neuralgia. It has also been used for other neuropathic pain conditions such as diabetic peripheral neuropathy and carpal tunnel syndrome. Recent studies have shown that it may be helpful in the treatment of low back pain and osteoarthritis.²⁵⁴ Lidocaine reduces

the ectopic transmission of pain signals sent to the dorsal horn of the spinal cord by inhibiting abnormal sodium channels in dermal nociceptors.¹² In chronic conditions such as arthritis, the joints are in a chronic inflammatory state resulting in upregulated sodium channels and sending aberrant pain signals to the spinal cord. Pain transmission may be interrupted by blocking these upregulated sodium channels.²⁵⁴

Topical menthol is widely available over the counter and has been used to relieve pain for millennia.¹³⁶ Although menthol is

generally well tolerated, it can induce skin irritation and cold hyperesthesia at higher doses. The analgesic effects of menthol are mediated through the TRP member-8 receptors, kappa-opioid receptors, and blockage of sodium channel and calcium influx.¹⁵⁶ Conversely, the pronociceptive and inflammatory effects are mediated via the TRP ankyrin-1 receptors. Topical menthol may produce skin irritation, but it is otherwise well tolerated.

Capsaicin, an active ingredient found in chili peppers, selectively activates the TRPV-1 receptors on nociceptive afferents.¹⁵ Capsaicin causes an initial burning sensation followed by desensitization via depletion of substance P and CGRP (pronociceptive neurotransmitters) in the afferent nerve terminal through capsaicin-induced activation of TRPV-1. Numerous studies have supported the use of daily low concentration (0.025 to 0.1%, available over the counter) of topical capsaicin for pain relief.^{13,15,72} High concentrations of capsaicin (e.g., 8% capsaicin patch [Qutenza]) lead to the long-term loss of function of nociceptors (dysfunction) via the ablation of TRPV-1-expressing afferent terminals.¹⁵¹ The higher concentration (8%) patch is FDA approved in the United States for the treatment of postherpetic neuralgia⁷² and diabetic peripheral neuropathy of the feet, but it has been used to treat other types of neuropathic pain. Pain relief produced with a single 30- to 60-minute application of 8% patch can last up to 3 months, but the application must be done in the office or

hospital setting to avoid unintentional contact with eyes, mucous membranes, respiratory tract, and skin.¹³

Therapy

While short-term rest and inactivity may be essential for the body to recover from an acute injury, many studies have shown prolonged rest is not beneficial for chronic pain conditions.^{167,227,271} Studies suggest that physical activity/exercise is an intervention with few adverse events that may improve pain severity, physical function, and consequent quality of life.^{182,246,292} Exercises can produce analgesic effects based on neuroplasticity changes on a central nervous system.²⁵³

Manual therapy has also been shown to help in the management of chronic pain. Manual therapy is a passive, purposeful movement applied by skilled clinicians to treat a painful area. Proposed mechanisms of this type of treatment include both peripheral (increasing pain threshold in peripheral pain receptors) and central (activation of descending pain inhibition).³⁴ Studies have also shown that activation of the autonomic nervous system produces a sympathetic excitatory state to reduce pain experience.²⁵¹ Table 38.13 outlines the various types of movement therapy, manual therapy, and modalities that can be utilized in the management of chronic pain.

TABLE 38.13 Commonly Used Movement Therapies and Treatment Modalities Used in Chronic Pain Management

Categories	Modalities	Description	Notes
Cognitive Modulation	Cognitive behavioral therapy (CBT)	CBT for chronic pain (CBT-CP) is about changing one's response to and relationship with pain so that it has a less deleterious impact on functioning and quality of life. ¹⁴³	CBT is considered the gold standard psychosocial intervention for chronic pain. ⁸²
	Hypnosis	The induction of a state of consciousness in which a person apparently loses the power of voluntary action and is highly responsive to suggestion or direction.	Research suggests hypnosis is efficacious for managing chronic pain. ⁵
Regenerative Therapy	Prolotherapy	Prolotherapy is a technique that involves the injection of an irritant, usually a hyperosmolar dextrose solution, typically in the treatment of chronic painful musculoskeletal conditions. ⁷⁷	Overall, prolotherapy remains a promising option for the treatment of painful musculoskeletal conditions, particularly when other standard treatments have proved ineffective. ⁷⁷
	Platelet-rich plasma (PRP)	PRP therapy is a form of regenerative medicine that harnesses and amplifies the natural growth factors found in our blood cells to help heal damaged tissue.	As of 2022, well-designed case control or cohort studies for the use of PRP have demonstrated efficacy in lumbar facet joint, lumbar epidural, and sacroiliac joint injections. PRP is utilized autogenously to facilitate healing and injection and has been studied in the long-term management of discogenic low back pain. In this regard, numerous studies have evaluated PRP to steroid injections in chronic pain states with favorable results. PRP represents an opportunity for a new strategy in the therapeutic treatment of degenerative states of spines, joints, and other locations throughout the body with evolving data demonstrating both safety and long-term efficacy. ¹⁰⁶
	Stem cells	Stem cell therapy for pain involves extracting, processing, and injecting stem cells into the affected area to promote tissue repair and regeneration. The stem cells release growth factors and other signaling molecules that help reduce inflammation, stimulate the formation of new blood vessels, and promote healing.	Stem cell therapy has shown promising results in the management of neuropathic, discogenic back, osteoarthritis, and musculoskeletal pain. ¹⁹¹

**TABLE
38.13****Commonly Used Movement Therapies and Treatment Modalities Used in Chronic Pain Management—cont'd**

Categories	Modalities	Description	Notes
Electrical Stimulation	Transcutaneous electrical nerve stimulation (TENS)	TENS is a method of pain relief involving the use of a mild electrical current. A TENS machine is a small, battery-operated device that has leads connected to sticky pads called electrodes.	There has been moderate-certainty evidence that pain intensity is lower during or immediately after TENS compared with placebo and without serious adverse events. ¹³⁴
	Cranial electrotherapy stimulation (CES)	CES uses a cellphone-sized device that stimulates the cranium and brain with a current that cannot usually be sensed by the consumer (<4 milliamps); additionally, the FDA has recognized CES as a class III device for treatment of depression, anxiety, and sleep disorders.	Research has shown CES to be highly effective in the management of chronic pain patients. It may be elevating the pain threshold due to its stress-reducing effects when anxiety and depression are reduced below clinical levels. ¹⁴⁰ However, there has also been evidence that although CES is an easily applicable and safe therapeutic option for chronic facial pain, active stimulation was not superior to sham stimulation. ¹⁹⁵
	Scrambler therapy	Scrambler therapy is a TENS that constitutes noninvasive cutaneous stimulation via skin electrodes and is approved by the FDA for pain treatment. ³	Scrambler therapy is a promising noninvasive, nonpharmacologic treatment and is a safe and effective neuromodulation modality that could be beneficial to treat many chronic pain syndromes, especially neuropathic pain. ³
	Biofeedback	Self-regulatory approach and therapeutic technique that uses various forms of auditory and visual physiologic monitoring to teach one to modify physiologic functions that are not normally under conscious control; common target responses include breathing rate, heart rate, blood pressure, skin temperature, and muscle tension.	It is concluded that biofeedback treatment can lead to improvements on various pain-related outcomes in the short and long terms, both as a standalone and as an adjunctive intervention. ²²⁹
Eastern Medicine	Acupuncture	A form of complementary medicine that involves pricking the skin or tissues with needles, used to alleviate pain and to treat various physical, mental, and emotional conditions; the technique is meant to encourage the movement of qi ("life energy") through the 14 channels (meridians) inside the body (as with acupressure; see later).	There is limited evidence that acupuncture is more effective than no treatment for chronic pain; and inconclusive evidence that acupuncture is more effective than placebo, sham acupuncture, or standard care. ⁸⁷
	Acupressure	The application of pressure or localized massage to specific sites on the body to control symptoms such as pain.	Studies have shown that auricular point acupressure is a promising pain management strategy that is not invasive and can be self-managed by participants for chronic low back pain with some credible evidence for more general pain relief. ⁵²
	Cupping	Cupping is a therapy in Chinese medicine in which heated glass cups are applied to the skin along the meridians of the body, creating suction as a way of stimulating the flow of energy.	The available evidence as of 2023 of very low to moderate quality suggests that cupping therapy is effective in managing chronic pain, knee osteoarthritis, low back pain, neck pain, chronic back pain, and herpes zoster. Moreover, it represents a promising, safe, and effective nonpharmacologic therapy that warrants wider application and promotion. ²⁵⁷
	Gua sha	Gua sha (pronounced "gwa-shah") is an East Asian healing technique. A trained professional uses a small, smooth-edged tool to firmly stroke the skin in one direction. This can help improve blood flow and reduce inflammation.	Gua sha increases microcirculation local to a treated area, and that increase in circulation may play a role in local and distal decrease in myalgia. ¹⁸⁹
Movement Therapy	Yoga	A Hindu spiritual and ascetic discipline, a part of which, including breath control, simple meditation, and the adoption of specific bodily postures, is widely practiced for health and relaxation.	Note: Yoga appears to be acceptable as a safe and self-manageable intervention for participants with musculoskeletal conditions. Further investigation of putative dose-response relationships between yoga and outcomes is warranted to determine optimum intervention parameters, requiring clear reporting of intervention parameters, home practice rates, and adherence. ²⁵⁸
	Pilates	A system of exercises using special apparatus, designed to improve physical strength, flexibility, and posture, and enhance mental awareness.	Numerous clinical trials reviewing use of Pilates as a rehabilitation tool have found it to be effective in achieving desired outcomes, particularly in the area of reducing pain and disability. ⁴²

Continued

TABLE 38.13**Commonly Used Movement Therapies and Treatment Modalities Used in Chronic Pain Management—cont'd**

Categories	Modalities	Description	Notes
Manual Therapies	Aquatic therapy	Gentler alternative to land-based physical therapy and serves various purposes. From injury recovery to athletic training, aquatic wellness pools can help people improve strength, flexibility, mobility, gait, and endurance in a relaxing setting.	The aquatic environment has broad rehabilitative potential, extending from the treatment of acute injuries through health maintenance in the face of chronic diseases, yet it remains an underused modality. There is an extensive research base supporting aquatic therapy, both within the basic science literature and clinical literature. Because of its wide margin of therapeutic safety and clinical adaptability, aquatic therapy is a very useful tool in the rehabilitative toolbox. ²⁴
	Tai chi	Involves a series of slow gentle movements and physical postures, a meditative state of mind, and controlled breathing.	There is some evidence that tai chi alone or as additional therapy with routine physical therapy may decrease pain and improve function disability for patients with low back pain, chronic osteoarthritis pain, and osteoporosis. ^{144,204}
	Chiropractor	A chiropractor is a licensed health care professional who treats disorders chiefly of the musculoskeletal system especially through the manual adjustment or manipulation of the spinal vertebrae to correct nervous system dysfunction.	Chiropractic care has been shown to help in managing or alleviating chronic pain without the use of drugs or surgery as well as being cost effective, considered safe, and satisfactory to a high number of patients. ¹⁹⁷
	Massage	Massage is the rubbing and kneading of muscles and joints of the body with the hands, especially to relieve tension or pain.	Massage therapists have the potential to further reduce suffering associated with chronic pain. ¹⁵¹
	Myofascial release (MFR)	MFR therapy is a type of gentle, constant massage that releases tightness and pain throughout the myofascial tissues.	The findings suggest that MFR can improve the effect of physical therapy alone and exercise therapy alone, and that MFR can be an effective adjuvant therapy. Meta-analysis showed that MFR has a significant effect on reducing back disability in patients with low back pain, but no significant effect on reducing pain intensity, improving quality of life, and improving lumbar range of motion. ⁵³
	Physical therapy (PT; stretching, strengthening, posture training, proprioception training)	PT, also known as physiotherapy, may include exercises, massages, and various treatments based on physical stimuli (e.g., heat, cold, electrical currents or ultrasound). The aim of PT is to relieve pain, help patients with mobility, strengthen weakened muscles, and show patients what they can do themselves to improve their own health.	After PT, pain-related scale scores were found to have improved, indicating benefits from PT. Additionally, pain severity and intensity, functional limitations, and disabilities were found to be reduced after 10 sessions of PT. ¹³⁸
Relaxation	Feldenkrais method (FM)	FM is a process that uses verbally and manually guided exploration of novel movements to improve individuals' self-awareness and coordination ²³⁸ ; these series of movements force the practitioner to use body sensation and perceptual feedback to choose between favorable (easy, comfortable) and unfavorable (painful, straining) positions.	Research clearly supports the effectiveness of FM for improvement of balance and chronic pain management. ²³⁸ Regarding elderly people, FM improves mobility, balance, and quality of life. Regarding pain, in people with chronic low back pain, FM has similar benefits such as back school lessons or core stability exercises; in people with cervical pain, the FM may be more adequate than conventional physical therapy. Regarding people with neurodegenerative diseases, the FM is effective in improving balance. ²⁷
	Progressive muscle relaxation	Teaches the patient to sequentially voluntarily contract and then relax various muscle groups; through this voluntary cycle, the patient gains insight into the sensation of muscle tension, which can then facilitate subsequent muscular relaxation.	Studies have indicated that progressive muscle relaxation therapy is a viable rehabilitative therapy for pain, fatigue, and stress symptoms in patients with fibromyalgia syndrome, ²⁶⁹ and relaxation training could be effective for patients with chronic musculoskeletal pain.
	Deep breathing	The act of deep breathing helps to activate the body's relaxation response, slowing heart rate and lowering blood pressure, which in turn can lead to a decrease in pain levels and an increase in overall well-being.	Paced breathing, at slow or normal frequencies, can reduce visceral pain intensity. Slow breathing increases heart rate variability and reduces anticipatory arousal. Effect of slow breathing on visceral pain is not mediated by autonomic or emotional responses but is moderated by pain catastrophizing. Applications of paced breathing in clinical conditions of visceral pain and effects of long-term practice warrant investigation. ¹⁰¹

Recently, virtual reality interventions have become more promising in the management of various pain conditions, including periprocedural, perioperative, and chronic pain.²⁵² Virtual reality is a simulated experience generated by using three-dimensional, computer-generated environments to give the user an immersive experience of the virtual world. There is much heterogeneity in the use of virtual reality interventions, such as frequency, duration, interaction, and type of software, so future long-term research is needed to establish a protocol for the use of this intervention.¹⁵⁷

Interventions

Interventional procedures can be helpful as an adjunct modality for chronic pain management (Table 38.14). Fluoroscopic interventions, such as epidural steroid injections, facet medial branch blocks,

and sacroiliac joint injections, have been shown to be safe interventions that are associated with good pain relief.¹²⁴ Multiple studies have shown effectiveness of radiofrequency ablation to denervate the facet joints in the treatment of axial low back pain.^{9,75,141,152,159} Pulsed radiofrequency ablation has also been used with success in the treatment of radicular pain, trigeminal neuralgia, occipital neuralgia, and shoulder and knee joint pain. Although many of these interventional procedures have been shown to be associated with significant pain relief, this relief may be temporary, and repeat procedures will be needed. For chronic intractable pain, neuromodulation utilizing implanted spinal cord stimulation, peripheral nerve stimulation, or dorsal root ganglion stimulation can be considered. Many studies have shown the benefits of these procedures,^{68,120,228} but due to the differences in their effectiveness, additional trials are necessary to evaluate the impact and timing of neuromodulation in the management of chronic pain.

TABLE 38.14 Summary of Interventional Procedures¹⁰⁹

Location	Treated Condition	Procedure Type	Description	Potential Complications
Axial/spine	Lumbar/cervical radiculopathy	Epidural steroid injection	A needle is placed in the epidural space through a foraminal (transforaminal) or midline (interlaminar) approach, or through the sacral hiatus (caudal) to administer medications such as glucocorticoids, local anesthetics, and/or saline. ^a	Epidural or soft tissue hematoma, epidural abscess, dural puncture, neural injury (spinal cord, nerve root). ^{124,160,165}
	Facet arthropathy	Facet injection	A needle is placed into the facet joint to administer medications such as glucocorticoids and local anesthetics. ^a	Postprocedure pain, nerve injury, dural puncture. ^{4,38,88,142,263,273}
		Medial branch block (MBB)	Diagnostic local anesthetic injections targeting the medial branches of the dorsal rami that supply sensory information from the affected facet joint(s). Relief is expected to last for the duration of the anesthetic and is required before performing radiofrequency ablation. ^a	Postprocedure pain, numbness, dysesthesia, nerve injury. ^{4,38,88,142,263}
		Medial branch radiofrequency ablation (RFA)	Performed after a successful MBB (>50–75% relief). In conventional RFA, a RF needle tip is placed parallel to the medial branch nerve that supplies the affected facet joint. The needle tip emits a high-frequency current to heat up surrounding tissues. Conventional RFA is typically performed with the needle tip set to 80°C for 90 seconds (variations in temperature and time exist). Pain relief can last 6–12 months. ^a	Postprocedure pain, numbness, dysesthesia, neuritis, nerve injury. ^{9,75,141,159}
	Sacroiliac (SI) pain	SI joint injection	A needle is placed into the SI joint to deliver glucocorticoids and local anesthetics. This can be done using anatomic landmark(s) or with image guidance (fluoroscopy, computed tomography, ultrasound).	Nerve injury, postprocedural pain. ^{128,231}
		Lateral branch block (LBB) and RFA	Treats extraarticular SI joint pain by targeting the sacral lateral branches of S1–S3 (additional levels may be involved). Relief is required from the LBB to progress to RFA. The pain relief can last 6–12 months. ^a	Postprocedure pain, numbness, dysesthesia, neuritis, nerve injury. ^{128,231}
	Malignant pain, severe, intractable nonmalignant pain, spasticity refractory to oral medications.	Intrathecal pump	The procedure is performed in the operating room under fluoroscopy. The patient is sedated and given prophylactic antibiotics. The catheter is placed into the intrathecal space to deliver medication directly to the spinal cord. The other end of the catheter is tunneled under the skin of the abdomen to connect to the infusion pump, which is usually placed in the anterior abdominal wall for easy access.	Injury to nerve roots or spinal cord, catheter malfunction, arachnoiditis, intrathecal abscess, meningitis, catheter-tip granuloma, postdural puncture headache. ^{2,69,70,274}

Continued

TABLE 38.14 Summary of Interventional Procedures—cont'd

Location	Treated Condition	Procedure Type	Description	Potential Complications
	Severe, refractory neuropathic pain, complex regional pain syndrome (CRPS), chronic radiculopathies, neuropathies, postherpetic neuralgia.	Neuromodulation	A 7–10-day trial with >50% improvement should be completed prior to implantation. The superior part of the lead contains electrodes that are placed in the epidural space depending on the location of the pain. The other end of the leads is tunneled subcutaneously to connect to the implantable pulse generator, which is usually located posteriorly, near the upper buttock.	Lead migration or fracture, spinal cord/root damage, epidural bleeding/infection, postdural headache in case of accidental intrathecal puncture. ^{92,147,262}
Peripheral	Arthritis, bursitis, tendinitis	Joint/bursa or peritendinous injection	Local anesthetic and glucocorticoid are injected into the joint space/bursa or around a tendon using anatomic landmark(s) and/or image guidance (ultrasound, fluoroscopy). Alternatively, hyaluronic acid derivative (US Food and Drug Administration approved for knee joint) can be used for joint injection. There are studies reporting that platelet-rich plasma (PRP) injections may help relieve pain associated with arthritis. ^{17,123} However, given heterogeneity in PRP preparation, dosing, and protocol, the evidence supporting PRP is limited.	Joint injury, tendon injury, increased pain. ^{17,123}
	Myofascial pain	Trigger point injection	A needle is inserted into trigger points, identified by palpation of taut bands that reproduce local and/or referred pain. Needling alone provides benefit, but it may also be done with injectates (local anesthetic, saline, botulinum toxin, dextrose, steroid, etc.). ^{63,186,205} A twitch response may be elicited with injection.	Increased pain, pneumothorax (trapezius injection). ^{10,235}
Head and neck	Occipital neuralgia, cluster headache, cervicogenic headache.	Occipital nerve block	A needle is directed toward the greater occipital nerve (more common) or lesser occipital nerve to deliver local anesthetic alone or with glucocorticoid; variations exist with injectates.	Puncture of occipital artery. ^{55,125}
	Trigeminal neuralgia, cancer pain of head and neck, postherpetic neuralgia.	Trigeminal (gasserian ganglion) block/denervation.	A needle is directed toward the trigeminal nerve to inject local anesthetic and glucocorticoid. Chemical denervation can also be done with phenol, alcohol, or glycerol injection. ^a	Dysesthesia, weakened masseter, facial asymmetry, hematoma, seizures, anesthesia dolorosa. ^{110,127}
Sympathetic chain	Upper extremity: CRPS, phantom limb pain, herpes zoster, peripheral nerve lesion, peripheral vascular disease, post-traumatic stress disorder.	Stellate ganglion block	A needle is directed toward the stellate ganglion (anterior portion of the transverse process of C7) guided by anatomic landmark(s), fluoroscopy, and/or ultrasound to deliver local anesthetic and/or steroid. Presence of Horner syndrome (ptosis, miosis, and anhidrosis) indicates a successful block.	Vascular injuries (carotid, internal jugular vein, and inferior thyroid artery), retropharyngeal hematoma, vagus nerve injury, brachial plexus injury, pneumothorax, neuraxial injection, seizures. ^{19,67}
	Lower extremity: CRPS, postherpetic neuralgia, phantom limb pain, peripheral vascular disease.	Lumbar sympathetic block	The spinal needle is directed toward the sympathetic chain at L2 and/or L3 vertebral bodies. Needle positioning is confirmed by contrast prior to injection of anesthetic and/or glucocorticoid. ^a	Intravascular injection, local anesthetic toxicity, injury to surrounding structures (the disk, genitofemoral nerve, kidney or ureter), retrograde ejaculation (bilateral injection). ^{19,67}
Upper abdomen	Acute or chronic abdominal pain from pancreatitis, or malignancies (pancreatic, gastric, biliary) mesenteric vascular occlusive disease.	Celiac plexus	The needle is directed toward the celiac plexus at the L1 vertebrae, confirmed by contrast under fluoroscopy. Local anesthetic is injected, with or without glucocorticoid. Neurolysis could be done with alcohol or phenol.	Orthostatic hypotension, diarrhea, nerve root injury, retroperitoneal bleeding, aortic dissection, sexual dysfunction. ^{16,83,206,266}

Note that bleeding, infection, vasovagal response, and/or injection site pain are common potential complications in all procedures listed.

If glucocorticoids are used, hyper/hypopigmentation, fat atrophy, and alopecia may occur at injection site, and systemic absorption is possible.

^aThese procedures are usually performed under fluoroscopic guidance.

While various medications and physical management modalities have been shown to be effective in the treatment of chronic pain, there is good evidence that personal beliefs about the body and pain play a powerful role in the behavior and emotional responses to pain.^{43,65,137,166} Exploration of individual beliefs, biases, and misconceptions about pain states are essential to the successful management of this chronic condition.

Cultural Biases and Misconceptions

Science and medicine have a long and disturbing history of reinforcing racist, antiethnic, and sexist attitudes and views, as well as ignoring and mistreating marginalized individuals. The science of pain is no exception. Misconceptions about pain sensitivity put forth by physicians and scientists from 1800 to 1965 reified existing social hierarchies.³⁹ It was believed that sensitivity to painful stimuli was linked to different personalities that were biologically defined by race, sex, or religion, and this reinforced existing racist and sexist science. Unfortunately, myths regarding pain and implicit biases persist, with some studies suggesting that sex stereotypes influence pain estimates due to beliefs that males are more stoic and females are more emotive.²⁷² This can contribute to an underestimation of pain in females. Racial misconceptions regarding heightened pain tolerance among Black people also continue to persist. For example, a recent study determined that White medical students and laypeople endorsed beliefs that there were biologic differences (i.e., having thicker skin) that accounted for higher pain tolerance in Black individuals, and these beliefs were linked with racial bias in pain treatment recommendations.¹¹⁹ Additionally, misconceptions and biases relating to age and development have influenced pain treatment and pain science. For example, until the 1980s, an erroneous belief that infants did not feel pain resulted in major surgeries being performed on infants without anesthesia.²¹¹ At the other end of the age spectrum, there has been an exclusion of adults over 65 years of age from studies evaluating pain therapies.⁴⁸ The disparities in pain management and quality of care are noteworthy. The undertreatment of pain in racial minority groups was brought to the forefront 3 decades ago²⁴⁵; nevertheless, pain care continues to be strongly influenced by pain biases and fallacies. An extensive body of research documents pain inequities by sociodemographic characteristics (e.g., racialized group, ethnicity, age, sex, sexual identity, socioeconomic position, regional location)¹²⁹ and by sexual minority status.¹ Tragically, in many instances, the individuals who experience the greatest burden from pain are the same individuals who are underrepresented in studies that aim to understand and alleviate pain.²⁴⁵ Thus recent calls to action have been put forth to make pain research more inclusive and to use antiracist principles in the conduct, reporting, and interpretation of pain research.^{121,154,179} In human studies, inclusion of representative groups by sex, age, race, ethnicity, among other forms of diversity is integral to gaining knowledge to address pain assessment and treatment inequities.¹⁹³ It is imperative that providers be mindful of these biases/misconceptions to provide optimal care. Researchers working to include representative groups in trials will allow for a continued, better understanding of pain.

Summary

The management of chronic pain is complex and necessitates multifaceted evaluations and a multidisciplinary team for effective treatment. Pain assessments should consider both biologic and

psychological factors contributing to the pain experience. Any primary physiologic cause of pain and coexisting medical and mental health conditions should be addressed. The most effective treatment plans typically combine medical management, physical (or movement) therapy, and CBT. Interventional procedures can also serve as adjunctive therapies to alleviate pain flare-ups. When designing a treatment plan, it is crucial to recognize that cultural biases and misconceptions may significantly impact its overall effectiveness.

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